

ON THE TANGENT BUNDLE AND THE DIVISOR THEORY OF A GENERAL MATROID

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ABSTRACT. Extending classical algebro-geometric constructions to arbitrary matroids, we construct a K -class $T_M \in K(M)$ for every loopless matroid M . When M is realizable by a linear subspace L , T_M recovers the K -class of the tangent bundle of the wonderful compactification W_L . We derive two formulas for the total Chern class of T_M (one combinatorial and one geometric) and show that the associated Todd class agrees with the Todd class appearing in the matroid Hirzebruch–Riemann–Roch formula. To develop a positivity theory entirely at the combinatorial level, we introduce the notion of “fake effective cone,” a combinatorial analogue of the classical effective cone, and use it to characterize big and nef divisors in $A(M)$. Finally, we define the β_S classes, obtained from Cremona conjugates of the classical α_S classes, and study their properties to provide a rich and computable family of combinatorially nef divisors.

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1. INTRODUCTION

A (loopless) matroid M on a finite ground set E carries rich geometric and combinatorial structure. Associated to M one has the Bergman fan Σ_M and its toric variety X_{Σ_M} , which we denote simply by X_M . Let $\mathbb{K}$ be a field. When M is realizable by a linear subspace $L \subset \mathbb{K}^E$, the De Concini–Procesi wonderful compactification W_L [DP95] provides a smooth projective model that sits naturally inside the permutohedral variety X_E , and the inclusion factors through X_M :

$$W_L \hookrightarrow X_M \hookrightarrow X_E.$$

In the realizable case the Chow and K -rings of X_M agree; geometric tools (Poincaré duality, Hodge–Riemann relations, Hirzebruch–Riemann–Roch, etc.) therefore transfer to the combinatorial setting.

Extending these geometric features beyond the realizable world is a central theme in recent work in matroid theory. Building on the foundational Hodge-theoretic breakthroughs of Adiprasito, Huh, and Katz [AHK18] and subsequent developments in matroid K -theory [EL23], much of the “projective geometry” of Chow rings admits a purely combinatorial incarnation. Nevertheless, while specific components of this picture have been extensively studied—such as the Todd class

of the permutohedral variety [CL21]—several natural geometric objects, most notably the tangent bundle and the associated Todd class for arbitrary (possibly nonrealizable) matroids, have not yet been given a fully satisfactory analogue. The principal aim of this paper is to fill that gap.

Below we summarize the main results; precise statements and proofs appear in the indicated references within the paper. As the matroid K -ring $K(X_M)$ admits standard λ -ring operations (analogous to the geometric setting), geometric constructions such as duals $(\cdot)^\vee$, exterior powers $\wedge^i$, and the rank function extend naturally to virtual classes such as our proposed tangent class.

Theorem 1.1 (Tangent class, Todd class, and Chow polynomial). *For every loopless matroid M there exists a canonical K -class $\tilde{T}_M \in K(X_E)$ whose restriction $T_M := i_M^* \tilde{T}_M \in K(X_M)$ (Definition 3.1) satisfies the following properties:*

- (1) Realizable compatibility. *If M is realizable by $L \subset \mathbb{K}^E$, then T_M coincides with the K -class of the tangent bundle of the wonderful compactification W_L .*
- (2) Chern class. *We provide two formulas for the Chern class of T_M (Theorem 3.4 and Theorem 4.1).*
- (3) Todd class and HRR. *The Todd class associated to T_M agrees with the Todd class appearing in Proposition 2.6, which formulates the Hirzebruch–Riemann–Roch theorem in the matroid setting. (See Corollary 3.20.)*
- (4) Chow polynomial via Euler characteristics. *The Euler characteristics of the exterior powers of the cotangent class $\Omega_M := T_M^\vee$ recover the coefficients of the Chow polynomial: for all i ,*

$$\dim A^i(M) = (-1)^i \chi(X_M, \wedge^i \Omega_M) = (-1)^i \deg(\mathrm{ch}(\wedge^i \Omega_M) \cdot \mathrm{td}(T_M)).$$

(See Theorem 3.23.)

Understanding the Todd class in the matroid Hirzebruch–Riemann–Roch formula is useful for computing Euler characteristics and for formulating vanishing statements in analogy with the Kawamata–Viehweg vanishing theorem (motivated by the positivity results in [EL23]). This motivates a study of big and nef divisors in the matroid setting.

Theorem 1.2 (Big and nef classes and the fake effective cone). *After comparing several candidate notions of nefness (and showing they are not equivalent; see Theorem 5.1), we introduce an operational notion of combinatorially big and nef divisors (Definition 5.7). Within this framework we prove the following results:*

- (1) Realizable compatibility. *If M is realizable by $L \subset \mathbb{K}^E$ and D is combinatorially big and nef for M , then the corresponding divisor on the wonderful compactification W_L is big and nef in the classical sense.*
- (2) Intersection inequalities. *We prove a collection of intersection-theoretic inequalities for nef and big-and-nef divisors.*
- (3) Fake effective cone. *There exists a combinatorial fake effective cone whose role is analogous to that of the classical effective cone in defining and detecting big and nef divisors (see Proposition 5.11).*
- (4) Matroid KV-type vanishing (rank 3). *We formulate a matroid analogue of the Kawamata–Viehweg vanishing statement, propose a weakened version, and verify the case for rank 3.*

A recurring practical question in the positivity theory is the following: There are many nef classes in the matroid Chow ring, but which of these can we effectively compute and use to test conjectures? The classical α_F -classes furnish important and highly computable examples, but they form a rather small, highly structured family and do not provide enough flexibility for exploration of the broader nef cone.

To produce a richer supply of controlled nef classes we study the β -classes, which are the Cremona conjugates of the α -classes on the permutohedral variety. The Cremona description is convenient conceptually, but the crucial point is that the β -classes give many new examples of divisors that are often nef and that can be computed in concrete cases. We summarize the results by:

Theorem 1.3 (Properties of the β -classes). *For a nonempty set $S \subseteq E$ we define the class β_S (Definition 2.5). The main properties are:*

- (1) *In X_E , α_S and β_S are Cremona conjugates of each other.*
- (2) *Let $[\mathcal{L}_S], [\mathcal{K}_S] \in K(X_M)$ be the line bundles corresponding to α_S, β_S . The exceptional isomorphism ζ_M (defined in Section 2.4) sends $-\mathcal{L}_S$ to $1 - \alpha_S$ and sends $\mathcal{K}_S$ to $1 + \beta_S$ (see Theorem 6.4).*
- (3) *Let M be a rank r matroid and let $S_1, \dots, S_{r-1} \subseteq E$ (repetitions allowed). Then*

$$\deg(\beta_{S_1} \cdots \beta_{S_{r-1}}) > 0 \iff \deg(\alpha_{S_1} \cdots \alpha_{S_{r-1}}) > 0.$$

- (4) *A weakened Kawamata–Viehweg vanishing statement holds for positive integral linear combinations of the α_S and β_S (see Corollary 6.5).*

Organization of the paper. Section 2 fixes notation and recalls necessary facts regarding the geometry and combinatorics of matroids. Section 3 defines the tangent and Todd classes, proving that the relevant functions are valutive in order to extend multiple results from realizable matroids to arbitrary ones. Section 4 computes the total Chern class via blow-ups and provides formulas for $c(T_M)$. Section 5 develops the positivity theory, introduces combinatorially big and nef divisors alongside the fake effective cone, and studies the Kawamata–Viehweg vanishing statement. Finally, Section 6 introduces and studies the properties of the β -classes. Importantly, Sections 5 and 6 are largely self-contained and can be read independently of Sections 3 and 4.

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2. PRELIMINARIES

Throughout this article, we assume that the reader is familiar with the main terminology in matroid theory. See [Oxl11] for general background on matroid theory. All matroids in the paper are assumed to be loopless unless otherwise mentioned. In this section, we establish the foundational background and notation used throughout the paper. We recall the standard algebraic and geometric frameworks for matroids—namely the wonderful compactification, the Chow ring, and the K -ring. We also review key operational tools required for our computations, including the valutive property and various morphisms such as pullback, pushforward, and deletion maps.

2.1. The wonderful variety. Let $E = \{1, \dots, n\}$ be the ground set, let $\mathbb{K}$ be a field, and let $L \subseteq \mathbb{K}^E$ be a linear subspace of dimension r that is not contained in any coordinate hyperplane. For $S \subseteq E$ set $L_S := L \cap \mathbb{K}^{E \setminus S}$, where $\mathbb{K}^{E \setminus S}$ denotes the coordinate subspace spanned by the coordinates indexed by $E \setminus S$. A subset $F \subseteq E$ is a *flat* if it is maximal among subsets yielding the same subspace L_F . The *wonderful compactification* W_L is obtained from $\mathbb{P}L$ by iteratively blowing up the linear subspaces $\mathbb{P}L_S$ corresponding to nontrivial flats, proceeding in increasing order of dimension and taking strict transforms at each step; see [DP95]. Equivalently, one may describe the construction as: first blow up all $\{\mathbb{P}L_S : \dim L_S = 1\}$ (points), then blow up the strict

transforms of the loci $\{\mathbb{P}L_S : \dim L_S = 2\}$ (lines), and so on. The result is a smooth projective variety of dimension $r - 1$, which we denote W_L .

2.2. Matroids and their geometric realizations. For $S \subseteq E$, we define $e_S = \sum_{i \in S} e_i$. Let M be a matroid on the ground set E , let $\mathcal{F} = (\emptyset \subsetneq F_1 \subsetneq \dots \subsetneq F_k \subsetneq E)$ be a flag of proper non-empty flats of M , and let $\rho_{\mathcal{F}}$ be the cone in $\mathbb{R}^E/\mathbb{R}(1, \dots, 1)$ generated by $\{e_F \mid F \in \mathcal{F}\}$.

Definition 2.1. The *Bergman fan* of a matroid M , denoted Σ_M , is the fan in $\mathbb{R}^E/\mathbb{R}(1, \dots, 1)$ whose cones are $\{\rho_{\mathcal{F}} \mid \mathcal{F} \text{ flag of flats of } M\}$.

In this way, a matroid M is associated with a toric variety X_{Σ_M} , which we will denote by X_M . The Boolean matroid U_n on the ground set $E = \{1, \dots, n\} = [n]$ is the matroid in which every subset is a flat. The toric variety X_E associated with U_n is called the *permutohedral variety*. For any matroid M on the ground set E , Σ_M is a subfan of Σ_{U_n} , and thus the toric variety X_M is an open subvariety of X_E . In particular, it is smooth.

For $L = \mathbb{K}^E$, the wonderful compactification W_L equals the toric variety X_E , and for any $L \subseteq \mathbb{K}^E$, the inclusion $L \hookrightarrow \mathbb{K}^E$ induces an inclusion $W_L \hookrightarrow X_E$.

We say that a matroid M is *realizable* or *realized* by $L \subseteq \mathbb{K}^E$ if the flats of M come from the flats of L . We say such L is a *realization* of M . For a realizable matroid M , the realization L is not unique, and the corresponding W_L could be different. Nonetheless, they share the same cohomology ring.

Proposition 2.2. [BHMPW22, Remark 2.13] *Let $L \subseteq \mathbb{K}^E$ be a realization of a matroid M . Then the inclusion $W_L \hookrightarrow X_E$ factors through X_M , and the pullback map*

$$A^*(M) \xrightarrow{\sim} A^*(W_L)$$

is an isomorphism. Here, $A^(M)$ denotes the Chow ring $A^*(X_M)$.*

This result extends to K -rings (the Grothendieck ring of vector bundles) as follows.

Proposition 2.3. [LLPP24, Proposition 1.6] *Let $L \subseteq \mathbb{K}^E$ be a realization of a matroid M . Then the restriction map*

$$K(M) \xrightarrow{\sim} K(W_L)$$

is an isomorphism. Here, $K(M)$ denotes the K -ring $K(X_M)$.

2.3. Chow ring of a matroid. The Chow ring $A^*(M)$ admits a presentation as a quotient of a polynomial ring [AHK18, Section 5.3]:

$$A^*(M) = \mathbb{Z}[x_F \mid F \text{ a nonempty proper flat of } M] / (I + J),$$

where:

- I is the ideal generated by products $x_F x_G$ for incomparable flats F and G ,
- J is the ideal generated by linear forms $\sum_{F \ni i} x_F - \sum_{F \ni j} x_F$ for each pair of elements $i, j \in E$.

When M is realized by L , the divisor $x_F \in A^*(W_L)$ is the exceptional divisor when blowing up at $\mathbb{P}L_F$ and then taking the pullbacks under further blow-ups. In general, X_M is not projective. The fascinating paper [AHK18] proved that the Chow ring of matroids satisfies several properties. In particular, the Chow ring vanishes in degree $\geq r$, and there is a degree map $\deg_M: A^{r-1}(M) \rightarrow \mathbb{Z}$ that is an isomorphism, mapping any monomial corresponding to a complete flag of flats to 1. The Chow ring also satisfies Poincaré duality in the sense that

$$A^i(M) \times A^{r-1-i}(M) \rightarrow A^{r-1}(M) \xrightarrow{\sim} \mathbb{Z}$$

is a perfect pairing for $0 \leq i \leq r - 1$, where the last map is given by $\deg_M$.

Definition 2.4. We define $\alpha = \alpha_i = \sum_{F \ni i} x_F \in A^1(M)$ and $\beta = \beta_i = \sum_{F \not\ni i} x_F \in A^1(M)$. Observe that the difference $\alpha_i - \alpha_j$ vanishes in the Chow ring, so α is independent of the element i we choose. Similarly, β is also independent of i .

For a set $\emptyset \subsetneq S \subset E$, we define

$$\alpha_S = \alpha - \sum_{\text{proper flats } F \supseteq S} x_F.$$

For example, $\alpha = \alpha_E$.

The following definition is new.

Definition 2.5. For a set $\emptyset \subsetneq S \subset E$, we define

$$\beta_S = \beta - \sum_{\text{proper flats } F \subseteq E \setminus S} x_F.$$

For example, $\beta = \beta_E$.

Note that $\alpha_S = \alpha_{\text{cl}(S)}$, where $\text{cl}(S)$ (the closure of S) is the smallest flat that contains S . In contrast, β_S cannot always be written as β_F for a flat F . We also write α_M or β_M for the α and β classes for the matroid M when there are multiple different matroids being discussed. We will try to make the context clear to avoid confusion between two potential meanings of the subscripts.

In the case of realizable matroids, α is the class of the vanishing loci of a generic section of the pullback of $\mathcal{O}(1)$ from the projective space $\mathbb{P}(\mathbb{K}^E)$ along the iterative blow-ups. Furthermore, α_F is the pullback of the hyperplane section that passes through the blow-up center corresponding to the flat F . (The class α_F is sometimes called h_F in other papers.)

For $E = \{1, \dots, n\}$, we have a natural birational morphism $X_E \rightarrow \mathbb{P}\mathbb{K}^E \cong \mathbb{P}^{n-1}$ given by the blow-up construction, where the blow-up centers are the intersections of coordinate hyperplanes $x_i = 0$. The Cremona involution

$$[t_1 : \dots : t_n] \rightarrow [t_1^{-1} : \dots : t_n^{-1}]$$

induces the Cremona involution $\text{crem} : X_E \rightarrow X_E$ (see [BEST23, Section 2.6]). The Cremona map exchanges x_I and $x_{E \setminus I}$. Therefore, in X_E we have $\beta_S = \text{crem}^*(\alpha_S)$ for any set $\emptyset \subsetneq S \subseteq E$. In particular, $\beta = \text{crem}^*(\alpha)$.

For any matroid M on the ground set E with the inclusion $i : X_M \rightarrow X_E$, the pullback map i^* sends x_F to x_F if F is a flat of M , and 0 otherwise. Therefore, for any set $\emptyset \subsetneq S \subseteq E$, α_S, β_S in M are the pullbacks of α_S, β_S in X_E , respectively.

2.4. K-theory and Chern character. Larson, Li, Payne, and Proudfoot introduced the K -ring of a general matroid in [LLPP24]. The Chern character map $\text{ch} : K(M) \rightarrow A^*(M) \otimes \mathbb{Q}$, which sends a line bundle $\mathcal{L}$ to $\exp(c_1(\mathcal{L}))$, induces a ring isomorphism over $\mathbb{Q}$ [Ful98, Example 15.2.16(b)]:

$$\text{ch} : K(M) \otimes \mathbb{Q} \xrightarrow{\sim} A^*(M) \otimes \mathbb{Q}.$$

Let $\mathcal{L}_F$ denote the line bundle associated with the divisor α_F . There is also a ring isomorphism [LLPP24, Theorem 1.8]

$$\zeta_M : K(M) \xrightarrow{\sim} A^*(M),$$

called the *exceptional isomorphism* sending

$$[\mathcal{L}_F] \mapsto \frac{1}{1 - \alpha_F} = 1 + \alpha_F + \alpha_F^2 + \dots.$$

It turns out that $[\mathcal{L}_F]$ generates the K -ring of M , and ζ_M is determined by the image $\zeta_M([\mathcal{L}_F])$.

In [LLPP24], the authors defined the Euler characteristic map $\chi : K(M) \rightarrow \mathbb{Z}$ by sending the class K to

$$(1) \quad \chi(K) = \deg_M (\zeta_M(K) \cdot (1 + \alpha + \alpha^2 + \dots)) \in \mathbb{Z},$$

which agrees with the Euler characteristic map when M is realizable. Using the Poincaré duality, one shows the following.

Proposition 2.6 (Matroid Hirzebruch–Riemann–Roch, cf. Section 3.1). *[EL23, Proposition 4.11] For a matroid M , there is a unique Todd class $\text{Todd}_M \in A^*(M)_{\mathbb{Q}}$ such that for $K \in K(M)$, we have*

$$\chi(K) = \deg_M (\text{ch}(K) \cdot \text{Todd}_M).$$

Moreover, the degree 0 part of Todd_M is 1.

The degree 0 part of Todd_M can be deduced by considering $K = \mathcal{L}_E^{\otimes k}$. However, to the author's knowledge, there is no known general formula to calculate Todd_M . We will provide a method to compute it via Theorem 4.1 and Corollary 3.20. The Todd class for the special case where $M = U_n$ was extensively studied in [CL21].

2.5. Maps between Chow ring of matroids. Let $M = (E, \mathcal{I})$ be a matroid and let $S \subseteq E$. We denote by M^S , $M \setminus S$, and M_S the restriction, deletion, and contraction, respectively.

Lemma 2.7 (Pullback map). *[BHMPW22, Section 2.6] Let M be a matroid of rank r , and let F be a nonempty proper flat of M . There is a unique graded algebra homomorphism*

$$\varphi_M^F : A^*(M) \longrightarrow A^*(M_F) \otimes A^*(M^F),$$

called the pullback map, such that for each flat G of M ,

$$\varphi_M^F(x_G) = \begin{cases} 0, & F \text{ and } G \text{ are incomparable,} \\ 1 \otimes x_G, & G \subsetneq F, \\ x_{G \setminus F} \otimes 1, & F \subsetneq G. \end{cases}$$

Moreover, φ_M^F is surjective and additionally satisfies

$$\begin{aligned} \varphi_M^F(x_F) &= -(1 \otimes \alpha_{M^F} + \beta_{M_F} \otimes 1), \\ \varphi_M^F(\alpha_M) &= \alpha_{M_F} \otimes 1, \\ \varphi_M^F(\beta_M) &= 1 \otimes \beta_{M^F}. \end{aligned}$$

There is also a pushforward map.

Lemma 2.8 (Pushforward map). *[BHMPW22, Section 2.6] Let M be a matroid of rank r , and let F be a nonempty proper flat of M . There is a unique graded algebra homomorphism*

$$\psi_M^F : A^*(M_F) \otimes A^*(M^F) \longrightarrow A^*(M),$$

called the pushforward map, that satisfies, for any collection $\mathcal{S}_1$ of proper flats of M strictly containing F and any collection $\mathcal{S}_2$ of nonempty proper flats of M strictly contained in F ,

$$\psi_M^F \left(\prod_{F' \in \mathcal{S}_1} x_{F' \setminus F} \otimes \prod_{F'' \in \mathcal{S}_2} x_{F''} \right) = x_F \prod_{F' \in \mathcal{S}_1} x_{F'} \prod_{F'' \in \mathcal{S}_2} x_{F''}.$$

The composition $\psi_M^F \circ \varphi_M^F$ is multiplication by the element x_F , and the composition $\varphi_M^F \circ \psi_M^F$ is multiplication by the element $\varphi_M^F(x_F)$. In particular, for $f \in A^{r-2}(M)$, we have $\deg_M(f \cdot x_F) = \deg_F(\varphi_M^F(f))$, where $\deg_F$ is defined as $\deg_{M_F} \otimes \deg_{M^F}$.

Definition 2.9 (Deletion map). [BHMPW22, Lemma 3.3] For a matroid M on the ground set E , and an element $i \in E$, there is a deletion map

$$\theta_i : A^*(M \setminus i) \rightarrow A^*(M), \quad x_F \mapsto x_F + x_{F \cup \{i\}}$$

where a variable in the target is set to zero if its label is not a flat of M .

Moreover, if i is not a coloop, then

$$\deg_{M \setminus i} = \deg_M \circ \theta_i.$$

If i is a coloop, then

$$\deg_{M \setminus i} = \deg_M \circ \alpha_M \circ \theta_i,$$

where the middle map in the composition denotes multiplication by α_M in the Chow ring of M .

2.6. Valutive Invariants. For a matroid M on the ground set E , the *matroid polytope* $P_M \subset \mathbb{R}^E$ is defined to be the convex hull of all e_B , where B is a basis of M .

A function f from the class of matroids on the ground set E to an abelian group is called *valuative* if it factors through the map that assigns to each matroid M the indicator function of its matroid polytope P_M . That is, for any matroids $M_1, \dots, M_k$ and integers $a_1, \dots, a_k$ such that $\sum a_i 1_{P_{M_i}} = 0$, we require that $\sum a_i f(P_{M_i}) = 0$.

Proposition 2.10. [EHL23, Corollary 7.9] or [DF10, Theorem 5.4] *For a matroid M , the indicator function of its matroid polytope P_M can be expressed as a linear combination of indicator functions of matroid polytopes of Schubert matroids of the same rank. Moreover, Schubert matroids are realizable over any infinite field.*

Remark 2.11. Although the original statement applies to all matroids (possibly having loops), if we assume M to be loopless, we can take every matroid in the linear combination to be loopless.

Therefore, if we show that a valutive function is zero for realizable matroids over $\mathbb{C}$, then it is zero for all matroids, and we can extend geometric and combinatorial results from realizable to non-realizable matroids.

3. TANGENT BUNDLE AND ITS CHERN CLASS

In this section, we define the cotangent and tangent classes of a general matroid by adapting the conormal exact sequence from classical algebraic geometry. We then use the tautological quotient class to compute explicit formulas for the total Chern class of the tangent bundle. Let X_E be the permutohedral variety associated with the ground set $E = \{1, \dots, n\}$. For a matroid M of rank r on E , [BEST23, Definition 3.9] defines the *tautological quotient K -class* $Q_M \in K(X_E)$. If M is realizable by $L \subset \mathbb{K}^E$, then Q_M is the class of a vector bundle Q_L of rank $n - r$ on X_E which admits a regular section whose vanishing locus is W_L . In this realizable case, Q_L is the normal bundle of W_L in X_E . Denote the inclusion by $i : W_L \hookrightarrow X_E$. The following definition is motivated by the conormal exact sequence for the ideal sheaf $\mathcal{I}$ of W_L in X_E :

$$0 \rightarrow \mathcal{I}/\mathcal{I}^2 \rightarrow i^* \Omega_{X_E} \rightarrow \Omega_{W_L} \rightarrow 0.$$

Definition 3.1. For an arbitrary matroid M , we define

$$\tilde{\Omega}_M := \Omega_{X_E} - Q_M^\vee \in K(X_E),$$

and its dual:

$$\tilde{T}_M := T_{X_E} - Q_M \in K(X_E).$$

In the realizable case, $i^* \tilde{\Omega}_M = \Omega_{W_L}$ and $i^* \tilde{T}_M = T_{W_L}$.

For an arbitrary matroid M , we write $i_M : X_M \hookrightarrow X_E$. We set $\Omega_M = i_M^*(\tilde{\Omega}_M)$ and $T_M = i_M^*(\tilde{T}_M)$, and call Ω_M and T_M the *cotangent bundle* and *tangent bundle* of M , respectively. Note that these are K -classes in $K(M)$.

Definition 3.2. For a matroid M of rank r and an integer $k > 0$, we define

$$S_{k,M} = \sum_{F \text{ is a rank } r-k \text{ flat}} x_F \in A^1(M).$$

The total Chern class of T_M is

$$c(T_M) = \frac{c(i_M^*(T_{X_E}))}{c(i_M^*(Q_M))}.$$

By [CLS11, Proposition 13.1.2], the Chern class of the tangent bundle T_{X_E} is given by $\prod_{\emptyset \subsetneq I \subsetneq E} (1 + x_I)$. Upon restriction to X_M , only the terms corresponding to flats of M survive. Therefore,

$$(2) \quad c(i_M^*(T_{X_E})) = \prod_{\text{proper flats } F} (1 + x_F) = \prod_{i=1}^{r-1} (1 + S_{i,M}),$$

where the latter equality holds because two flats of the same rank are always incomparable.

The Chern class of $i_M^*(Q_M)$ is given as follows.

Theorem 3.3. [BEST23, Appendix 3] or [AL24, Section 4.3] For a matroid M of rank r , the total Chern class of $i_M^*(Q_M)$ is given by

$$\prod_{i=0}^{r-1} \frac{1}{(1 + \alpha - \sum_{j=1}^i S_{j,M})}.$$

As a result, we have the following.

Theorem 3.4. For a matroid M of rank r , the total Chern class of its tangent K -class T_M is given by

$$c(T_M) = \left(\prod_{i=1}^{r-1} (1 + S_{i,M}) \right) \cdot \left(\prod_{i=0}^{r-1} (1 + \alpha - \sum_{j=1}^i S_{j,M}) \right).$$

Corollary 3.5. For a matroid M of rank r , the total Chern class $c(T_M)$ lies in $\mathbb{Z}[\alpha_M, S_{1,M}, \dots, S_{r-1,M}]$.

Remark 3.6. In fact, for a rank r matroid M , we have

$$(1 + S_{r-1,M})(1 + \alpha - S_{1,M} - \dots - S_{r-1,M}) = (1 + \alpha - S_{1,M} - \dots - S_{r-2,M}).$$

So $c(T_M)$ lies in $\mathbb{Z}[\alpha_M, S_{1,M}, \dots, S_{r-2,M}]$, and we do not need $S_{r-1,M}$.

3.1. Hirzebruch-Riemann-Roch formula. Our strategy in this subsection is to use the valuative property of Euler characteristics to extend the classical Hirzebruch-Riemann-Roch formula from realizable matroids to arbitrary loopless matroids.

The Todd class of a vector bundle $\mathcal{E}$ on X is given by

$$\text{td}(\mathcal{E}) = \prod_i \frac{\alpha_i}{1 - e^{-\alpha_i}} \in A^*(X) \otimes \mathbb{Q},$$

where $\{\alpha_i\}$ are the Chern roots of $\mathcal{E}$. The Todd class can be given explicitly as a formal power series in the Chern classes:

$$\text{td}(\mathcal{E}) = 1 + \frac{c_1(\mathcal{E})}{2} + \frac{c_1(\mathcal{E})^2 + c_2(\mathcal{E})}{12} + \dots,$$

so the Todd class can be defined for a K -class.

For a smooth projective variety X and any coherent sheaf $\mathcal{F}$, the Hirzebruch-Riemann-Roch formula implies

$$\chi(\mathcal{F}) = \deg(\text{ch}(\mathcal{F}) \cdot \text{td}(T_X)),$$

where T_X is the tangent bundle. Our goal is to relate this formula to Proposition 2.6 (Matroid Hirzebruch-Riemann-Roch).

When M is realizable by L , the Todd class in Proposition 2.6 then equals $\text{td}(T_M)$. In this section, we will prove that this is true for arbitrary matroids. Our strategy is to use valuativity to extend from realizable to general matroids, and Proposition 3.7 is the key step.

Proposition 3.7. *Let X_E be the permutohedral variety. Fix an integer $r > 0$, a polynomial $z \in \mathbb{Q}[x, y_1, y_2, \dots, y_{r-1}]$, and a class $A \in A^*(X_E)$. For a rank r matroid M with ground set E , we set*

$$z_M = z(\alpha_M, S_{1,M}, \dots, S_{r-1,M}) \in A^*(M).$$

Define

$$\Phi_{z,A} : \{\text{rank } r \text{ Matroids on } E\} \longrightarrow \mathbb{Q}, \quad M \longmapsto \deg_M(i_M^*(A) \cdot z_M).$$

Then the map $\Phi_{z,A}$ is valutive for the set of rank r matroids.

The proof of Proposition 3.7 will be completed in Subsection 3.3.

3.2. Intersection numbers and flag counting functions. To prove Proposition 3.7, we must show that intersection numbers in the Chow ring behave valuably. Our strategy in this subsection is to express products of the divisors α, β , and x_F entirely in terms of combinatorial flag counting functions. Because these counting functions are known to be valutive, this will establish the valuativity of the intersection numbers. Recall that in the Chow ring $A^*(M)$, the classes α and β can be represented as sums of generators. For any fixed element $i \in E$, these classes satisfy the following identities:

$$(3) \quad \alpha = \sum_{F \ni i} x_F, \quad \beta = \sum_{F \not\ni i} x_F.$$

Proposition 3.8. *Let F be a flat of rank d .*

- (1) *For any $i \notin F$, we have $x_F \cdot \alpha = \sum_{G \supsetneq F, i \in G} x_F x_G$.*
- (2) *For any $i \in F$, we have $x_F \cdot \beta = \sum_{G \subsetneq F, i \notin G} x_F x_G$.*

Proof. For (1), substituting the definition of α from (3) using an element $i \notin F$, we have $x_F \alpha = x_F (\sum_{G \ni i} x_G)$. The product $x_F x_G$ is non-zero only if $F \subseteq G$ or $G \subseteq F$. Since $i \in G$ and $i \notin F$, the case $G \subseteq F$ is impossible. Moreover, G cannot be F itself. Thus, the only surviving terms are those where $G \supsetneq F$. The proof for (2) is symmetric, choosing $i \in F$ and observing that $G \subsetneq F$ are the only comparable flats not containing i . $\square$

Lemma 3.9. *Suppose $f = \prod_{F \in I} x_F$ is a degree d monomial, where I is a multiset of flats. If all flats in I are strictly between two flats $F_1 \subset F_2$ with $\text{rk}(F_1) = d_1$ and $\text{rk}(F_2) = d_2$, then $x_{F_1} f x_{F_2} = 0$ if $d_2 - d_1 \leq d$. (By convention we allow $F_1 = \emptyset$ or $F_2 = E$ and set $x_\emptyset = x_E = 1$.)*

Proof. Consider the matroid minor bounded by F_1 and F_2 (that is, contracting on F_1 and restricting to F_2). It is a rank $d_2 - d_1$ matroid, and everything with degree $\geq d_2 - d_1$ will vanish in the Chow ring. By the pullback map (Lemma 2.7) of F_1 and F_2 we get $f = 0$. $\square$

Lemma 3.10 (Basic properties of α and β). *If F is a flat of rank d , then $x_F \alpha^{r-d} = x_F \beta^d = 0$. Furthermore, $\deg(\alpha^{r-1}) = 1$ and $\deg(\beta^{r-1}) > 0$.*

Proof. By Proposition 3.8, multiplying x_F by α (resp. β) produces a sum of products $x_F x_G$ where $\text{rk}(G) > d$ (resp. $\text{rk}(G) < d$). Iterating this $r - d$ (resp. d) times forces the resulting flags to reach rank r (resp. 0), and the products vanish.

For the degree, Proposition 3.8 implies α^{r-1} is a sum over maximal flags. Since for any flat F and element $i \notin F$, there is a unique flat G of rank $\text{rk}(F) + 1$ containing $F \cup \{i\}$, the iteration yields a unique maximal chain, hence $\deg(\alpha^{r-1}) = 1$. In contrast, for a flat F and an element $i \in F$, there exists at least one flat G of rank $\text{rk}(F) - 1$ covered by F such that $i \notin G$, hence $\deg(\beta^{r-1}) > 0$. $\square$

When M is realizable, we can interpret this combinatorial result geometrically. α is the pullback of $\mathcal{O}(1)$, as a result, α^{r-d} can avoid everything of codimension $\geq d$, and the product $x_F \alpha^{r-d} = 0$ when $\text{rk}(F) \geq d$. Moreover, $\deg(\alpha^{r-1}) = 1$. The geometric meaning of β is not that straightforward, but it behaves in many ways like the dual of α .

Note that $\beta_M = S_{1,M} + \dots + S_{r-1,M} - \alpha_M \in \mathbb{Q}[\alpha_M, S_{1,M}, \dots, S_{r-1,M}]$, so we can handle β_M in our argument. For an integer d , denote $[d] = \{1, 2, \dots, d\}$

Definition 3.11. For a rank r matroid M on the ground set E , denote $\mathcal{H}$ as the set of flags of flats. For a subset $I = \{i_1, \dots, i_k\} \subseteq [r-1]$ write $\mathcal{H}_I$ for the set of flags of flats $\mathcal{G} = G_1 \subsetneq G_2 \subsetneq \dots \subsetneq G_k$ with $\text{rk}(G_t) = i_t$ for all t .

Let $\mathcal{F} = F_1 \subsetneq \dots \subsetneq F_l$ be a flag of flats, and let $I \subset \{1, \dots, r-1\}$. We define the number

$$N_{\mathcal{F}, I} = \#\{\mathcal{G} \in \mathcal{H}_I \text{ and } \mathcal{G} \sqcup \mathcal{F} \in \mathcal{H}\},$$

where the notation $\mathcal{G} \sqcup \mathcal{F} \in \mathcal{H}$ also implies $\mathcal{G}$ and $\mathcal{F}$ are disjoint.

For an element $s \in E$ in the ground set, we define

$$N_{\mathcal{F}, I, s} = \#\{\mathcal{G} \in \mathcal{H}_I \mid \mathcal{G} = (G_1 \subsetneq \dots \subsetneq G_k), s \notin G_k, \text{ and } \mathcal{G} \sqcup \mathcal{F} \in \mathcal{H}\}.$$

Lemma 3.12. Let M be a matroid of rank r on a ground set E , and let d be an integer such that $0 < d < r$. Suppose that $\mathcal{F} = \emptyset$ is the empty flag. Then for any element $s \in E$, we have

$$\sum_{\substack{I \subseteq [d] \\ d \in I}} (-1)^{|I|} N_{\mathcal{F}, I, s} = \sum_{I \subseteq [d]} (-1)^{|I|} N_{\mathcal{F}, I}.$$

Proof. We proceed by induction on d . When $d = 1$, there is exactly one flat of rank 1 containing s . This corresponds to the empty set $I = \emptyset$ in the right-hand-side, so the formula is correct.

To compute the left-hand-side, we first sum over all G_k , then subtract the case when G_k contains s . For the part summing over all possible G_k , the value is $\sum_{I \subseteq [d], d \in I} (-1)^{|I|} N_{\mathcal{F}, I}$.

For a set $I = \{i_1, \dots, i_k\} \subseteq \{1, \dots, r-1\}$ with $i_k = d$, and a number $j \leq k$, we define the number

$$N_{\mathcal{F}, I, j} = \#\{\mathcal{H}_I \ni \mathcal{G} = G_1 \subsetneq G_2 \subsetneq \dots \subsetneq G_k, s \in G_{j+1}, s \notin G_j, \text{ and } \mathcal{G} \sqcup \mathcal{F} \in \mathcal{H}\}.$$

In particular, it is allowed to have $j = 0$ and $G_0 = \emptyset$.

We have

$$\sum_{\substack{I \subseteq [d], d \in I \\ 0 \leq j < |I|}} (-1)^{|I|} N_{\mathcal{F}, I, s} = \sum_{\substack{I \subseteq [d], d \in I}} (-1)^{|I|} N_{\mathcal{F}, I} - \sum_{\substack{I \subseteq [d], d \in I \\ 0 \leq j < |I|}} (-1)^{|I|} N_{\mathcal{F}, I, j}.$$

Denote $I = \{i_1, \dots, i_k\}$. Note that $s \notin G_j$ and $s \in G_{j+1}$ is equivalent to $G_{j+1} \supseteq (G_j \cup \{s\})$, and there is a unique flat of rank $i_j + 1$ containing $G_j \cup \{s\}$. Therefore, if $i_{j+1} > i_j + 1$,

$$N_{\mathcal{F}, \{i_1, \dots, i_k\}, j} = N_{\mathcal{F}, \{i_1, \dots, i_j, i_j+1, i_{j+1}, \dots, i_k\}, j}.$$

All terms will cancel out in the summation of $N_{\mathcal{F}, I, j}$ except for the case $i_{k-1} + 1 = i_k$ and $j = k-1$. Therefore,

$$\sum_{\substack{I \subseteq [d], d \in I \\ 0 \leq j < |I|}} (-1)^{|I|} N_{\mathcal{F}, I, j} = \sum_{\substack{I = \{i_1, \dots, i_k\} \subseteq [d] \\ d-1, d \in I, \text{ and } j=k-1}} (-1)^{|I|} N_{\mathcal{F}, I, j},$$

which is equivalent to $\sum_{I \subseteq [d-1], d-1 \in I} (-1)^{|I|+1} N_{\mathcal{F}, I, s}$.

By induction, this is $-\sum_{I \subseteq [d-1]} (-1)^{|I|} N_{\mathcal{F}, I}$, and we are done. $\square$

Lemma 3.13. Let M be a rank r matroid on the ground set E , and let $0 \leq d < r$. We have

$$\deg_M(\alpha^{r-1-d} \beta^d) = (-1)^d \sum_{I \subseteq [d]} (-1)^{|I|} N_{\emptyset, I},$$

where $\emptyset$ is the empty flag.

Proof. We prove this by induction on d . This is true for $d = 0$ as both sides are 1.

Expand one of the β 's. Choose an $s \in E$, the terms contributing to the degree should be

$$\sum_{\substack{s \notin F, \text{ and} \\ F \text{ has rank } d}} x_F \beta^{d-1} \alpha^{r-d-1}.$$

By the pullback map and pushforward map in 2.7 and 2.8 with respect to the flat F ,

$$\deg_M(x_F \beta^{d-1} \alpha^{r-d-1}) = \deg_F(\varphi_M^F(\beta^{d-1} \alpha^{r-d-1})) = \deg_{M_F}(\beta^{d-1}).$$

Sum over all rank d flats that do not contain s . By induction and Lemma 3.12, the summation is

$$(-1)(-1)^{d-1} \sum_{I \subseteq [d], d \in I} (-1)^{|I|} N_{\emptyset, I, s} = (-1)^d \sum_{I \subseteq [d]} (-1)^{|I|} N_{\emptyset, I}.$$

□

Theorem 3.14 (Flag-valuations). *[FS24, Theorem 6.2] Let $F_1 \subsetneq \dots \subsetneq F_k$ be any fixed flag of subsets of E . Define*

$$\widehat{\Phi}_{F_1, \dots, F_k} : \{\text{Matroids on } E\} \longrightarrow \mathbb{Z}$$

by

$$(4) \quad \widehat{\Phi}_{F_1, \dots, F_k}(M) = \begin{cases} 1, & F_i \text{ is a flat for all } 1 \leq i \leq k, \\ 0, & \text{otherwise.} \end{cases}$$

Then $\widehat{\Phi}_{F_1, \dots, F_k}$ is valutive.

Moreover, for any rank-vector $(r_1, \dots, r_k) \in \mathbb{Z}^k$, define

$$\widehat{\Phi}_{F_1, \dots, F_k}^{r_1, \dots, r_k} : \{\text{Matroids on } E\} \longrightarrow \mathbb{Z}$$

by

$$(5) \quad \widehat{\Phi}_{F_1, \dots, F_k}^{r_1, \dots, r_k}(M) = \begin{cases} 1, & F_i \text{ is a flat and } \text{rk}(F_i) = r_i \\ & \text{for all } 1 \leq i \leq k, \\ 0, & \text{otherwise.} \end{cases}$$

Then $\widehat{\Phi}_{F_1, \dots, F_k}^{r_1, \dots, r_k}$ is also valutive.

Corollary 3.15. *Let $\mathcal{F} = F_1 \subsetneq \dots \subsetneq F_k$ be any fixed flag of subsets of E and $I \subseteq \{1, \dots, |E| - 1\}$. Define*

$$\widehat{N}_{\mathcal{F}, I} : \{\text{Matroids on } E\} \longrightarrow \mathbb{Z}$$

by

$$(6) \quad \widehat{N}_{\mathcal{F}, I}(M) = \begin{cases} N_{\mathcal{F}, I}, & \mathcal{F} \text{ is a flag of flats in } M \\ 0, & \text{otherwise.} \end{cases}$$

Then $\widehat{N}_{\mathcal{F}, I}$ is valutive.

Proof. $\widehat{N}_{\mathcal{F}, I}$ is the sum of the functions in (5) enumerating over all possible flats and rank. □

Proposition 3.16. *Let M be a matroid of rank r , and let*

$$\emptyset = F_0 \subsetneq F_1 \subsetneq \dots \subsetneq F_k \subsetneq F_{k+1} = E$$

be a flag of flats with $\text{rk}(F_i) = r_i$. Let $d_0, d_1, \dots, d_{k+1}$ be nonnegative integers summing to $r - 1$, such that $d_i > 0$ for all $1 \leq i \leq k$. Consider the monomial

$$m = \beta^{d_0} x_{F_1}^{d_1} \dots x_{F_k}^{d_k} \alpha^{d_{k+1}} \in A^{r-1}(M).$$

Setting $\mathcal{F} = (F_1, \dots, F_k)$, we can express the degree of m as

$$\deg_M(m) = \sum_I c_I N_{\mathcal{F}, I},$$

where the coefficients $c_I \in \mathbb{Z}$ depend only on the sequences of exponents (d_j) and ranks (r_j) .

Proof. Apply the pullback map $\varphi_M^{F_1}: A^*(M) \rightarrow A^*(M_{F_1}) \otimes A^*(M^{F_1})$ from Lemma 2.7. By Lemma 2.8, the degree $\deg_M(m)$ factors as the product of the degrees in the respective Chow rings. For $0 \leq i \leq k$, we denote by M_i the minor of M bounded by F_i and F_{i+1} (that is, the minor obtained by contracting F_i and restricting to F_{i+1}).

In the matroid $M^{F_1} = M_0$, the only nontrivial contribution comes from the terms β and $x_{F_1}^{d_1-1}$. We expand the pullback $\varphi_M^{F_1}(x_{F_1}^{d_1-1}) = (-1 \otimes \alpha_{M^{F_1}} - \beta_{M^{F_1}} \otimes 1)^{d_1-1}$ using the binomial theorem. Because M^{F_1} has rank r_1 , only the term with degree $r_1 - 1$ in $A^*(M^{F_1})$ has a non-vanishing degree map. Since we already have a factor of β^{d_0} , the unique term providing the correct degree is

$$(-1)^{d_1-1} \binom{d_1-1}{r_1-d_0-1} \deg_{M_0}(\alpha_{M_0}^{r_1-d_0-1} \beta_{M_0}^{d_0}),$$

and the corresponding tensor factor in M_{F_1} is $\beta_{M_{F_1}}^{d_1-(r_1-d_0)} x_{F_2}^{d_2} \dots x_{F_k}^{d_k} \alpha_{M_{F_1}}^{d_{k+1}}$.

Iterating this pullback process along the chain of flats $F_1, \dots, F_k$ yields

$$\deg_M(\beta^{d_0} x_{F_1}^{d_1} \dots x_{F_k}^{d_k} \alpha^{d_{k+1}}) = C \prod_{i=0}^k \deg_{M_i}(\alpha_{M_i}^{t_i} \beta_{M_i}^{s_i}),$$

for integers t_i, s_i and a combinatorial coefficient C that depend only on the exponents d_j and the ranks r_j (but not on the specific matroid M).

By Lemma 3.13, each factor $\deg_{M_i}(\alpha_{M_i}^{t_i} \beta_{M_i}^{s_i})$ is a $\mathbb{Z}$ -linear combination of $N_{\mathcal{F}_i, I}$ associated with the two-flat flag $\mathcal{F}_i = (F_i, F_{i+1})$ (where $I \subseteq \{r_i + 1, \dots, r_{i+1} - 1\}$). Consequently, the product $C \prod_{i=0}^k \deg_{M_i}(\alpha_{M_i}^{t_i} \beta_{M_i}^{s_i})$ expands into a linear combination of products $\prod_{i=0}^k N_{\mathcal{F}_i, I_i}$. Since each such product canonically equals $N_{\mathcal{F}, I}$ for the full flag $\mathcal{F} = (F_1, \dots, F_k)$ with index set $I = \bigcup_i I_i$, we conclude that $\deg_M(m)$ is a $\mathbb{Z}$ -linear combination of $N_{\mathcal{F}, I}$, with coefficients depending exclusively on the d_j and r_j . $\square$

Remark 3.17. The proof may be shortened by applying [Eur20, Theorem 3.2].

3.3. Valuativity. With the combinatorial flag counting identities established, we are now ready to prove Proposition 3.7 and, subsequently, the full Matroid Hirzebruch-Riemann-Roch formula. Recall that i_M is the inclusion $X_M \rightarrow X_E$.

Proof of Proposition 3.7. The pullback map i_M^* between Chow rings sends x_F to x_F if F is a flat of M , and 0 otherwise.

We can assume that A and z are monomials, and we expand z_M as the sum of $x_{F_1}^{d_1} \dots x_{F_k}^{d_k} \alpha^{d_{k+1}}$ over all flags of flats having the same sequence of ranks. This is the same as enumerating the flats in the definition of $N_{\mathcal{F}, I}$, where the flag of flats $\mathcal{F}$ is given by A . By Proposition 3.16, $\Phi_{z, A}$ is a linear combination of $\widehat{N}_{\mathcal{F}, I}$, and thus is valutive. $\square$

Proposition 3.18. *For a matroid M of rank r , the Todd class of the tangent bundle $\text{td}(T_M)$ lies in $\mathbb{Q}[\alpha_M, S_{1, M}, \dots, S_{r-2, M}]$.*

Proof. The Chern class of T_M lies in $\mathbb{Z}[\alpha_M, S_{1, M}, \dots, S_{r-2, M}]$. $\square$

We have the following lemma.

Lemma 3.19. [LLPP24, Lemma 6.4] *For any class $R \in K(X_E)$, the map $M \mapsto \chi(i_M^* R)$ is valutive.*

The restriction map $i_M^* : K(X_E) \rightarrow K(M)$ is surjective. For a rank r matroid M and a $K \in K(M)$, we can write $K = i_M^*(R)$ for some $R \in K(X_E)$. For a fixed $R \in K(X_E)$, we would like to show that both sides of the equation

$$\chi(i_M^*(R)) = \deg_M(\text{ch}(i_M^*(R)) \cdot \text{td}(T_M))$$

are valutive.

Since $i_M : X_M \rightarrow X_E$ is an open immersion, the Chern class, hence the Chern character map commutes with the pullback map i_M^* (see [Ful98, Theorem 3.2(d)]). Hence, $\text{ch}(i_M^*(K)) = i_M^*(\text{ch}(K))$. Since $\text{td}(T_M) \in \mathbb{Q}[\alpha, S_{1,M}, \dots, S_{r-1,M}]$, the right-hand side is valutive by Proposition 3.7.

The equation holds for realizable matroids of the same rank, so it can be extended to arbitrary matroids by Proposition 2.10. In particular,

$$\chi(K) = \deg_M(\text{ch}(K) \cdot \text{td}(T_M)).$$

Corollary 3.20. *For a matroid M of rank r , the Todd class $\text{Todd}_M = \text{td}(T_M)$.*

Remark 3.21. The factors in Theorem 3.4 are not the Chern roots since T_M has rank $r - 1$, not $2r - 1$. Remarkably, the Todd class still equals

$$\prod_{i=1}^{2r-1} \frac{t_i}{1 - e^{-t_i}},$$

where $t_i = S_{i,M}$ for $0 < i < r$ and $t_{r+i} = \alpha - \sum_{j=1}^i S_{j,M}$ for $0 \leq i < r$. This is because the Todd class formula does not depend on the rank of the bundle.

3.4. Serre duality for matroids. In [LLPP24], the Serre duality equation $\chi(\mathcal{E}) = (-1)^{r-1} \chi(\mathcal{E}^\vee \otimes \omega_M)$ is proved by extending the equation from the geometric case. We show the result from a different perspective.

Corollary 3.22. *Let M be a matroid of rank r , and let $\omega_M \in K(M)$ denote the line bundle with first Chern class*

$$c_1(\omega_M) := -c_1(T_M) = -r\alpha + \sum_{i=1}^{r-2} (r-i-1)S_{i,M}.$$

Then for $\mathcal{E} \in K(M)$ we have

$$\chi(\mathcal{E}) = (-1)^{r-1} \chi(\mathcal{E}^\vee \otimes \omega_M).$$

Proof. First, recall a general fact about Todd classes: for any vector bundle V with Chern roots x_i , $\text{td}(V) = \prod \frac{x_i}{1 - e^{-x_i}}$ and $\exp(-c_1(V)) = \prod e^{-x_i}$. Multiplying these together yields

$$\text{td}(V) \cdot \exp(-c_1(V)) = \prod \frac{x_i}{e^{x_i} - 1} = \text{td}(V^\vee) = \text{td}(V)^\vee.$$

Applying this to the tangent bundle T_M and noting that $c_1(\omega_M) = -c_1(T_M)$, we obtain the relation $\text{td}(T_M) \exp(c_1(\omega_M)) = \text{td}(T_M)^\vee$.

Substituting this identity into the Hirzebruch–Riemann–Roch formula

$$\chi(\mathcal{E}^\vee \otimes \omega_M) = \deg_M(\text{ch}(\mathcal{E}^\vee) \cdot \text{ch}(\omega_M) \cdot \text{td}(T_M)),$$

and using the relations $\text{ch}(\mathcal{E}^\vee) = \text{ch}(\mathcal{E})^\vee$ and $\text{ch}(\omega_M) = \exp(c_1(\omega_M))$, this becomes:

$$\chi(\mathcal{E}^\vee \otimes \omega_M) = \deg_M(\text{ch}(\mathcal{E})^\vee \cdot \text{td}(T_M)^\vee) = \deg_M((\text{ch}(\mathcal{E}) \cdot \text{td}(T_M))^\vee).$$

The dual operation $(\cdot)^\vee$ acts on the degree k component of the Chow ring by multiplication by $(-1)^k$. Since the degree map $\deg_M$ extracts the top-degree component in $A^{r-1}(M)$, evaluating the dual of the product $\text{ch}(\mathcal{E}) \cdot \text{td}(T_M)$ pulls out a global sign of $(-1)^{r-1}$. Therefore,

$$\chi(\mathcal{E}^\vee \otimes \omega_M) = (-1)^{r-1} \deg_M(\text{ch}(\mathcal{E}) \cdot \text{td}(T_M)) = (-1)^{r-1} \chi(\mathcal{E}).$$

□

3.5. The Chow (Poincaré) polynomial. For a matroid M realized by L over $\mathbb{C}$, W_L is a wonderful compactification obtained by an iterated sequence of blow-ups along smooth subvarieties, each of which is again a wonderful compactification of smaller rank. By inductively assuming that each blow-up center satisfies $h^{p,q} = 0$ for all $p \neq q$, it follows that the final compactification W_L also satisfies:

$$h^{p,q}(W_L) = 0 \quad \text{for all } p \neq q,$$

i.e., its Hodge diamond is supported purely on the diagonal. Furthermore, the cycle map from the Chow ring to the cohomology ring continues to be an isomorphism. (For example, see [Voi02, Theorem 7.31].)

For a matroid M of rank r , one defines the Chow polynomial

$$P_M(t) = \sum_{p=0}^{r-1} \dim A^p(M) t^p.$$

If M is realized by L , we have

$$\dim A^p(M) = \dim H^{2p}(W_L) = h^{p,p}(W_L) = (-1)^p \chi(W_L, \Omega_{W_L}^p).$$

For an arbitrary matroid M of a fixed rank r , we can expand $(-1)^p \chi(\Omega_M^p)$ by $\deg(\text{ch}(\Omega_M^p) \cdot \text{td}(T_M))$. The Chern class of Ω_M^p can be derived from the Chern class of the tangent bundle, so it lies in the algebra generated by $\alpha, S_{1,M}, \dots, S_{r-2,M}$. In particular, its Euler characteristic function is a polynomial in $\alpha, S_{1,M}, \dots, S_{r-1,M}$, and is valutive. Moreover, the Chow polynomial is also valutive [FS24, Section 8.4]. Since the equation $\dim A^p(M) = (-1)^p \chi(\Omega_M^p)$ holds for realizable matroids, it holds for arbitrary matroids.

Theorem 3.23. *The coefficients of the Chow polynomial are given by the formula:*

$$\dim A^p(M) = (-1)^p \chi(\Omega_M^p) = (-1)^p \deg(\text{ch}(\Omega_M^p) \cdot \text{td}(T_M)).$$

This shows how the dimension of the Chow ring can be derived from the tangent bundle class. For example, applying Theorem 3.23 at $p = 1$ to a rank r matroid M yields:

$$-\dim A^1(M) = \chi(T_M^\vee).$$

The dimension of $A^1(M)$ equals the number of generators minus the number of linear relations, which is the number of proper flats minus the number of rank 1 flats, plus 1 (cf. Section 2.3).

Furthermore, recall from Definition 3.1 that:

$$\chi(T_M^\vee) = \chi(i_M^*(T_{X_E})^\vee) - \chi(i_M^*(Q_M)^\vee).$$

We can compute the first term, $\chi(i_M^*(T_{X_E})^\vee)$, using the following theorem.

Theorem 3.24. *Let M be a matroid on ground set E , and let F be a proper flat of M . Let $[\mathcal{X}_F] \in K(M)$ denote the line bundle with $c_1([\mathcal{X}_F]) = x_F$. Then:*

$$\chi(-[\mathcal{X}_F]) = \chi(\mathcal{O}(-x_F)) = 0.$$

In particular, letting $n = |E|$, we have:

$$\chi(i_M^*(T_{X_E})^\vee) = n - 1 - (\text{number of proper flats}).$$

Proof. If M is realizable by L , we can treat x_F as a divisor on W_L . This yields the following short exact sequence of sheaves on W_L :

$$0 \longrightarrow \mathcal{O}(-x_F) \longrightarrow \mathcal{O} \longrightarrow \mathcal{O}_{x_F} \longrightarrow 0.$$

Because W_L and the vanishing set of x_F are rational varieties, we have $\chi(\mathcal{O}) = \chi(\mathcal{O}_{x_F}) = 1$ in this geometric setting. It immediately follows that $\chi(\mathcal{O}(-x_F)) = 0$.

Next, fix a proper subset $\emptyset \subsetneq S \subsetneq E$ and let $[\mathcal{X}_S] \in K(X_E)$ denote the line bundle with $c_1([\mathcal{X}_S]) = x_S$. Consider the function on the set of matroids on a ground set E :

$$f(M) = \chi(i_M^*(-[\mathcal{X}_S])),$$

which is valuative by Lemma 3.19. If S is not a flat of M , then $i_M^*(-[\mathcal{X}_S]) = 0$ and $f(M) = 1$. If S is a flat and M is realizable, then $f(M) = 0$ as shown above.

The difference $f - \mathbb{1}_{\{S \text{ is not a flat}\}}$ is also valuative and vanishes on all realizable matroids by Theorem 3.14; therefore, it must vanish on all matroids. Thus, $\chi(-[\mathcal{X}_F]) = 0$.

Recall that the Chern character of a class in the K -ring depends only on its total Chern class and its rank. The fact that the rank of $i_M^*(T_{X_E})^\vee$ is $n-1$ can be verified in the realizable case and extended to arbitrary matroids via a short valuative argument. By (2), we have:

$$c(i_M^*(T_{X_E})^\vee) = \prod_{\text{proper flats } F} (1 - x_F).$$

Therefore, the Chern characters

$$\text{ch}(i_M^*(T_{X_E})^\vee) \quad \text{and} \quad \text{ch}\left(\bigoplus_{\text{proper flats } F} -[\mathcal{X}_F]\right) = \sum_{\text{proper flats } F} \text{ch}(-[\mathcal{X}_F])$$

differ only in their degree-0 components, which represent the ranks of their respective classes. In particular, $\text{ch}(i_M^*(T_{X_E})^\vee) - (n-1) = \sum_{\text{proper flats } F} (\text{ch}(-[\mathcal{X}_F]) - 1)$. Hence,

$$\begin{aligned} \chi(i_M^*(T_{X_E})^\vee) &= \deg(\text{ch}(i_M^*(T_{X_E})^\vee) \cdot \text{td}(T_M)) \\ &= \left(n-1 - \sum_{\text{proper flats } F} 1\right) \deg(\text{td}(T_M)) + \sum_{\text{proper flats } F} \deg(\text{ch}(-[\mathcal{X}_F]) \cdot \text{td}(T_M)) \\ &= n-1 - \sum_{\text{proper flats } F} 1, \end{aligned}$$

where we use that $\deg(\text{td}(T_M)) = 1$ and $\deg(\text{ch}(-[\mathcal{X}_F]) \cdot \text{td}(T_M)) = \chi(-[\mathcal{X}_F]) = 0$. This completes the proof. $\square$

This result directly yields the following corollary.

Corollary 3.25. *Let M be a matroid on ground set E . We have:*

$$\chi(i_M^*(Q_M)^\vee) = |E| - (\text{number of rank 1 flats}).$$

4. A GEOMETRIC DERIVATION VIA BLOW-UPS

In the previous section, we derived a product formula for $c(T_M)$. We now present an alternative geometric derivation. This approach begins with the realizable case, mirroring the blow-up construction of the De Concini-Procesi wonderful compactification, and then extends to all matroids via valuativity. While more computationally intensive, this method provides direct geometric insight and yields a recursive formula for the same Chern class, as stated in the following theorem.

Theorem 4.1. *For each integer $n \geq 0$, there is a polynomial $T_n(x, y_1, \dots, y_{n-1}) \in \mathbb{Z}[x, y_1, \dots, y_{n-1}] \subset \mathbb{Z}[x, y_1, \dots]$ such that $T_0 = 1$, and for $n > 0$,*

$$T_n = (1+x)^{n+1} + \sum_{\substack{1 \leq j \leq n-1, \\ 0 \leq i \leq n+1-j}} \binom{n+1-j}{i} (T_{j-1}) ((1+y_j)(1-y_j)^i - 1) \left(x - \sum_{k=1}^{j-1} y_k\right)^{n+1-i-j}.$$

For an arbitrary matroid M of rank r , we write $T_{r-1}(M) = T_{r-1}(\alpha_M, S_{1,M}, \dots, S_{r-2,M})$. Then for all matroids M , the total Chern class of the tangent bundle is given by

$$c(T_M) = T_{r-1}(M).$$

Remark 4.2. For small values of r , one can verify that this recursive formula expands to the product formula given in Theorem 3.4. This also explains why there is no $S_{r-1,M}$ term, because the corresponding blow-up centers have codimension 1.

The remainder of this section is dedicated to proving Theorem 4.1.

4.1. The realizable case. Suppose M is a rank r matroid realized by $L \subseteq \mathbb{K}^E$. The wonderful compactification W_L is obtained by successively blowing up $\mathbb{P}^{r-1}$ along all linear subspaces corresponding to flats. At each step, we track the effect on Chern classes.

For any smooth variety X , we write $c(X)$ for the total Chern class of the tangent bundle $c(T_X)$.

Fix smooth projective varieties X and Y with $i : X \hookrightarrow Y$. Let d be the codimension of X . When we blow up Y at center X , there is a blow-up diagram

$$(7) \quad \begin{array}{ccc} \tilde{X} & \xhookrightarrow{j} & \tilde{Y} \\ g \downarrow & & \downarrow f \\ X & \xhookrightarrow{i} & Y \end{array}$$

Let $N = N_{X/Y}$ denote the normal bundle of X in Y (rank d). Then the exceptional divisor $\tilde{X} \subset \tilde{Y}$ is isomorphic to $\mathbb{P}(N)$, and the normal bundle $N_{\tilde{X}/\tilde{Y}}$ is $\mathcal{O}_{\mathbb{P}(N)}(-1)$.

Proposition 4.3. [Ful98, Theorem 15.4] With the above notation, and $\zeta = c_1(\mathcal{O}_{\mathbb{P}(N)}(1))$,

$$c(\tilde{Y}) - f^*(c(Y)) = j_*(g^*c(X) \cdot \gamma),$$

where

$$\gamma = \frac{1}{\zeta} \left[\sum_{i=0}^d g^*c_{d-i}(N) - (1 - \zeta) \sum_{i=0}^d (1 + \zeta)^i g^*c_{d-i}(N) \right].$$

Here, the polynomial expression inside the brackets has no constant term with respect to ζ , making this formal division $\frac{1}{\zeta}$ well-defined.

In order to compute $c(W_L)$ by the formula above, we need to compute the normal bundle of the blow-up center during the blow-up process. We will use the following Proposition.

Proposition 4.4. [Ful98, B.6.10] Suppose $X \subsetneq Y \subsetneq Z$ is a sequence of closed regular embeddings. Let $\tilde{Z} = \text{Bl}_X Z$ and let $\tilde{Y} = \text{Bl}_X Y$ be the strict transform of Y . Then the embedding of $\tilde{Y}$ into $\tilde{Z}$ is a regular embedding, and its normal bundle is given by:

$$N_{\tilde{Y}/\tilde{Z}} = \pi^* N_{Y/Z} \otimes \mathcal{O}(-E),$$

where E is the exceptional divisor of the blow-up in $\tilde{Y}$ and π is the projection map $\tilde{Y} \rightarrow Y$.

Let's compute $c(W_L)$ for a realization $L \subset \mathbb{K}^E$ of small dimension to motivate the main theorem.

Example 4.5. Suppose M is a rank 3 matroid realized by $L \subset \mathbb{K}^E$. To construct W_L we start from $\mathbb{P}^L \cong \mathbb{P}^2$. Every rank 2 flat F corresponds to a point p_F , and W_L is the blow-up of $\mathbb{P}^2$ at all points p_F for F a rank 2 flat.

The total Chern class of $\mathbb{P}^n$ equals $(1 + \alpha)^{n+1}$, so for $\mathbb{P}^2$ we have $c_1 = 3\alpha$ and $c_2 = 3\alpha^2$. Blowing up $\mathbb{P}^2$ at p_F contributes $-x_F$ to c_1 and $-x_F^2$ to c_2 . Therefore,

$$c_1(W_L) = 3\alpha - \sum_{F \text{ rank } 2} x_F = 3\alpha - S_{1,M}, \quad c_2(W_L) = 3\alpha^2 - \sum_{F \text{ rank } 2} x_F^2 = 3\alpha^2 - S_{1,M}^2.$$

4.2. Chern class of wonderful compactification. We compute the Chern class throughout the blow-up construction. The key steps involve tracking the normal bundles using Proposition 4.4 and relating the exceptional divisors via Lemma 4.6.

Write $W_L = X^{r-2} \rightarrow \dots \rightarrow X^1 \rightarrow X^0 = \mathbb{P}^{r-1}$, where the map $X^{i+1} \rightarrow X^i$ blows up the strict transforms of the linear subspaces $\mathbb{P}L_F$ of dimension i (equivalently, flats of rank $r-i-1$). Consider blowing up a rank $r-i$ flat F during the blow-up $X^i \rightarrow X^{i-1}$. Denote the blow-up center by X_F . By Proposition 4.3, the difference of the Chern classes is given by

$$c(\tilde{X}) - f^*(c(X^{i-1})) = j_*(g^*c(X_F) \cdot \gamma),$$

where $\tilde{X}$ is the result of the blow-up at X_F , and the notations of the maps coincide with the blow-up diagram (7).

The blow-up center X_F is itself a wonderful compactification for the contracted matroid M_F , which has rank i . Its flats correspond to the flats $G \supseteq F$ of M , so we understand $c(X_F)$.

Initially, $N_{\mathbb{P}^{i-1}/\mathbb{P}^{r-1}} = \mathcal{O}(1)^{\oplus(r-i)}$. Applying Proposition 4.4 iteratively shows that the normal bundle $N = N_{X_F/X^{i-1}} = \mathcal{O}(\alpha_{M_F})^{\oplus(r-i)} \otimes \mathcal{O}(-\sum_{F \subsetneq G \subsetneq E} x_G)$, where x_G denotes the exceptional divisor coming from blowing-up G . Equivalently,

$$N_{X_F/X^{i-1}} = \mathcal{O}(\alpha_{M_F} - \sum_{F \subsetneq G \subsetneq E} x_G)^{\oplus(r-i)}.$$

We have described the Chern class of the normal bundle of X_F in $A^*(X_F)$. To apply Proposition 4.3, we need to know $j_*g^*c(N)$. One might expect that for a flat $G \not\supseteq F$, the corresponding Chern class in $A^*(X_F)$ should be related to the corresponding Chern class in W_L . The following lemma explains their relation.

Lemma 4.6. *Let $X \subsetneq Y \subsetneq Z$ be smooth varieties. Let $\pi : Z' = \text{Bl}_X Z \rightarrow Z$ denote the blow-up of Z along X , and let $Y' = \text{Bl}_X Y \subset Z'$ be the strict transform of Y . Denote by $E_X^Y \subset Y'$ the exceptional divisor of the blow-up $Y' \rightarrow Y$, and $E_X^Z \subset Z'$ the exceptional divisor of the blow-up $Z' \rightarrow Z$. Consider the blow-up $\tilde{Z} = \text{Bl}_{Y'} Z'$, with exceptional divisor $E_{Y'} \subset \tilde{Z}$. Let $j : E_{Y'} \hookrightarrow \tilde{Z}$ be the inclusion, and $g : E_{Y'} \rightarrow Y'$ the projection. Notice that the following diagram coincides with the blow-up diagram (7):*

$$\begin{array}{ccc} E_{Y'} & \xhookrightarrow{j} & \tilde{Z} \\ g \downarrow & & \downarrow f \\ Y' = \text{Bl}_X Y & \xhookrightarrow{i} & Z' = \text{Bl}_X Z \end{array}$$

Then, in the Chow group $A_*(E_{Y'})$, we have:

$$g^*(E_X^Y) = j^*f^*(E_X^Z).$$

Proof. Observe that $Y' \subset Z'$ and the exceptional divisor $E_X^Z \subset Z'$ intersect transversely, and

$$Y' \cap E_X^Z = E_X^Y.$$

Since Y' meets E_X^Z transversely, the strict transform of E_X^Z under f meets the exceptional divisor $E_{Y'}$ along the preimage of E_X^Y . Hence $j^*f^*(E_X^Z) = g^*(E_X^Y)$. $\square$

Here, $Y' = X_F$ and $Z' = X^{i-1}$. Hence for a monomial $t = \alpha_{M_F}^{t_0} S_{1,M_F}^{t_1} \dots S_{i-2,M_F}^{t_{i-2}} \in A^*(X_F)$, we have

$$g^*(t) = j^*(\alpha_M^{t_0} S_{1,M}^{t_1} \dots S_{i-2,M}^{t_{i-2}}).$$

Finally, note that $j^*(-x_F) = \mathcal{O}_{\mathbb{P}(N)}(-1) = -\zeta$. By the projection formula, we have

$$j_*(\zeta^l g^*(t)) = x_F(-x_F)^l \alpha_M^{t_0} S_{1,M}^{t_1} \dots S_{i-2,M}^{t_{i-2}}.$$

At the stage $X^i \rightarrow X^{i-1}$, the centers corresponding to flats of rank $r-i$ are pairwise incomparable (and hence disjoint), so they may be blown up simultaneously. The summation of the $(-1)^l(x_F)^{l+1}$ then becomes $(-1)^l S_{i,M}^{l+1}$.

To see exactly how the formula in Theorem 4.1 emerges from Proposition 4.3, we translate the geometric data of the blow-up into our polynomial variables. Let $n = r - 1$. At the j -th stage of the construction (where we blow up the centers corresponding to flats), the normal bundle N has rank $d = n + 1 - j$.

The first Chern class of the associated line bundle is represented by $x - \sum_{k=1}^{j-1} y_k$. Therefore, the total Chern class of the normal bundle is $c(N) = (1 + x - \sum_{k=1}^{j-1} y_k)^{n+1-j}$. Extracting the $(d-i)$ -th Chern class of N directly yields the binomial term in our sum:

$$g^* c_{d-i}(N) = \binom{n+1-j}{i} \left(x - \sum_{k=1}^{j-1} y_k \right)^{n+1-i-j}.$$

The other terms in the summand map directly to the remaining components of Proposition 4.3:

- The recursive factor T_{j-1} corresponds to $g^* c(X_F)$, the Chern class of the blow-up center.
- The algebraic factor $((1 + y_j)(1 - y_j)^i - 1)$ is the result of applying the pushforward j_* to the ζ terms in γ , which converts the formal expressions involving the exceptional divisor class ζ into our polynomial variable y_j .

Summing these contributions over all blow-up stages j and indices i recovers the recursive formula. Therefore, Theorem 4.1 holds for realizable matroids.

To complete the proof, we must show that this formula holds for all matroids. For a fixed class $A \in A^*(X_E)$, both the function $M \mapsto \deg_M(i_M^*(A) \cdot T_{r-1}(M))$ and the function $M \mapsto \deg_M(i_M^*(A) \cdot c(T_M))$ are valutive by Proposition 3.7. Since the functions are equal on realizable matroids, we obtain

$$\deg(i_M^*(A) \cdot T_{r-1}(M)) = \deg(i_M^*(A) \cdot c(T_M))$$

for *all* matroids M and *all* $A \in A^*(X_E)$. By Poincaré duality and the fact that i_M^* is surjective, we conclude that $T_{r-1}(M) = c(T_M)$ for all matroids M , completing the proof.

5. BIG AND NEF DIVISORS

5.1. Nef divisors. In classical algebraic geometry, nefness is naturally defined via intersection numbers with curves. Because arbitrary matroids lack projective embeddings, we must define nefness combinatorially. We compare three increasingly weaker conditions for a divisor to be considered nef in the following theorem.

Theorem 5.1. *Let M be a matroid of rank r and write*

$$l = \sum_F l_F x_F \in A^1(M)$$

for a degree-1 class (a divisor). Consider the following properties of l :

- (P1) *l is the pullback of a nef divisor on X_E ; i.e., there exists $\tilde{l} \in A^1(X_E)$ with $\tilde{l}$ nef and $i_M^*(\tilde{l}) = l$.*
- (P2) *For every flag $\mathcal{F} = (F_1 \subsetneq \dots \subsetneq F_k)$ of proper flats (including the empty flag) there is an expression $l = \sum_F c_F x_F$ with rational coefficients such that $c_{F_i} = 0$ for all i and $c_F \geq 0$ for every flat F .*
- (P3) *For every flag $\mathcal{F} = (F_1 \subsetneq \dots \subsetneq F_k)$ of proper flats (including the empty flag) there is an expression $l = \sum_F c_F x_F$ with rational coefficients such that $c_{F_i} = 0$ for all i and $c_F \geq 0$ for every flat F with the property that $\mathcal{F} \sqcup \{F\}$ is again a flag (i.e., F can be inserted into the flag).*

Then (P1) $\implies$ (P2) $\implies$ (P3), and both implications are strict.

Proof. If $\tilde{l} = \sum_S c_S x_S$ is nef on X_E then by [Lar24, Proposition 4.4.3] its coefficients satisfy the positivity property required in (P2); pulling back to $A^*(M)$ preserves that property, so (P1) $\Rightarrow$ (P2). Clearly (P2) $\Rightarrow$ (P3).

We give brief counterexamples to show the implications are strict.

(P3) $\nRightarrow$ (P2). Let $M = U_{3,6}$. Choose

$$l = x_1 + x_2 + 2x_{12} + x_{14} + x_{25} + x_{16} + x_{26}.$$

One checks directly (finite case check) that l satisfies (P3), but the coefficient of x_{14} cannot be set to 0 without producing a negative coefficient elsewhere.

(P2) $\nRightarrow$ (P1). Let $M = U_{3,4}$ and take

$$l = 2\alpha - x_{23} - x_{24} - x_{13} - x_{14} = x_1 + x_2 + 2x_{12} = x_3 + x_4 + 2x_{34}.$$

Then l satisfies (P2): for any flag, at least one of the two forms $x_1 + x_2 + 2x_{12}$ and $x_3 + x_4 + x_{34}$ avoids the flag (Conceptually, the divisor is “basepoint free”.) However l is not the pullback of a nef divisor on X_E : a nef divisor $\tilde{c} = \sum_S c_S x_S$ on X_E must satisfy the submodularity relations $c_{I \cup J} + c_{I \cap J} \leq c_I + c_J$ with $c_\emptyset = c_E = 0$ by convention (see [BES24, Proposition 2.2.6]). Checking these inequalities for the lift of l leads to a contradiction (one finds some $c_T < 0$), so (P2) does not imply (P1). $\square$

Definition 5.2. A class $l \in A^1(M)$ is *combinatorially nef* if it satisfies (P3). It is *combinatorially ample* if, moreover, for each flag, one can arrange the representing coefficients to be strictly positive on the divisors that extend the flag. When the meaning is clear we drop the adjective “combinatorially” and simply say “nef” or “ample”.

Lemma 5.3. [AHK18, Proposition 4.5] *The combinatorially nef and combinatorially ample classes form nonempty convex cones in $A^1(M)_{\mathbb{R}}$, and the ample cone is the interior of the nef cone.*

Example 5.4. For any matroid M and any nonempty $S \subseteq E$ the classes α_S and β_S are pullbacks from X_E ; in particular they satisfy (P1) and hence are nef.

The following proposition explains the geometric meaning of (P2) and (P3) in the realizable case.

Proposition 5.5. *Let M be realizable by $L \subset \mathbb{K}^E$ and let $l \in A^1(M)$ be a divisor.*

- (1) *If l satisfies (P2) then l is semiample on W_L (a positive multiple is basepoint free).*
- (2) *If l satisfies (P3) then l is nef on W_L .*

Proof. (1) If l admits, for every flag $\mathcal{F}$, an expression with nonnegative coefficients vanishing on the flag, then we can choose a nonnegative rational representative of l that avoids any point of W_L lying in the intersection of the divisors corresponding to $\mathcal{F}$. The union of these intersections of divisors is equal to W_L , and by finiteness of the set of flags this implies a positive multiple of l is basepoint free.

- (2) If $C \subset W_L$ is an irreducible curve then C is contained in the intersection of divisors corresponding to some flag $\mathcal{F}$; by (P3) we can ensure that the coefficients of l on divisors containing that intersection are nonnegative, so $l \cdot C \geq 0$.

$\square$

Remark 5.6. For an arbitrary (and possibly noncomplete) fan Δ , Gibney and Maclagan [GM12] defined three analogous cones of divisors $G_\Delta \subseteq L_\Delta \subseteq F_\Delta$ for the toric variety corresponding to Δ , establishing that the inclusions between them are strict in general. In relation to Theorem 5.1, the condition (P1) corresponds to membership in G_Δ , while (P3) corresponds to L_Δ .

5.2. Big and nef divisors. In algebraic geometry, the following four statements are equivalent (and characterize big and nef divisors).

- (1) D is nef, and it can (rationally) be written as ample + effective.
- (2) There is an effective divisor Z such that, for all k sufficiently large, $D - Z/k$ is ample.
- (3) D is nef, and $\deg(D^{r-1}) > 0$.
- (4) D is nef and in the interior of the effective cone.

So far, we can define big and nef divisors for matroids by the third statement.

Definition 5.7. For a matroid M of rank r , a combinatorially nef divisor l is *combinatorially big and nef* if $\deg(l^{r-1}) > 0$. Again, we omit the notion combinatorially if the meaning is clear.

It is tempting to define the *combinatorially effective cone* for matroids, so that the big and nef divisors can be characterized by the equivalence of the following.

Goal 5.8 (the definition of “combinatorially effective” will have these properties be equivalent).

- (1) D is nef, and it can (rationally) be written as ample + *combinatorially effective*.
- (2) There is a *combinatorially effective* divisor Z such that, for all k sufficiently large, $D - Z/k$ is ample.
- (3) D is nef, and $\deg(D^{r-1}) > 0$.
- (4) D is nef and in the interior of the *combinatorially effective cone*.

For the geometric background and intuition, we refer to [Laz04, Section 2.2]

Example 5.9. The cone of divisors generated by nonnegative sums of x_F s, is not the correct one.

Consider the matroid $U_{3,n}$ for $n > 4$. The class $D = 2\alpha - x_{12} - x_{34} = \alpha_{12} + \alpha_{34}$ is nef, and $\deg(D^2) = 2^2 - 1 - 1 = 2$. However, x_i for $i > 4$ cannot appear. Here, if we pick $E = x_5$, then $D - \epsilon E$ cannot be written as a nonnegative sum of x_F for any positive ϵ . Thus, (3) does not imply (4) in this setting.

Here is a proposal to make such a definition.

Definition 5.10. Let M be a matroid of rank r . The *fake effective cone* $\mathcal{F}_M$ is the closed convex cone of classes

$$\mathcal{F}_M = \{ D \in A^1(M)_{\mathbb{R}} : \deg(D \cdot \ell_1 \cdots \ell_{r-2}) \geq 0 \text{ for all combinatorially nef } \ell_1, \dots, \ell_{r-2} \}.$$

A class $D \in \mathcal{F}_M$ is called *fake effective*.

Theorem 5.11. *With the fake effective cone $\mathcal{F}_M$ in place, the four properties (1)–(4) in Goal 5.8 are equivalent: for a divisor D the following are equivalent.*

- (1) D is nef, and it can (rationally) be written as $D = A + Z$ with A ample and $Z \in \mathcal{F}_M$.
- (2) There exists $Z \in \mathcal{F}_M$ such that $D - \frac{1}{k}Z$ is ample for all sufficiently large integers k .
- (3) D is combinatorially nef and $\deg(D^{r-1}) > 0$.
- (4) D is combinatorially nef and lies in the interior of $\mathcal{F}_M$.

To prove this, we will apply a version of the reverse Khovanskii-Tessier inequality, and we will need the notion of Lorentzian polynomials. For definition and properties on Lorentzian polynomials, we refer to [BH20].

Proposition 5.12. [AHK18, Theorem 8.9] *Let M be a matroid of rank r , and $\ell_1, \dots, \ell_n$ be combinatorially ample divisors. Then the function*

$$f(x_1, \dots, x_n) = \frac{1}{(r-1)!} \deg((x_1 \ell_1 + \cdots + x_n \ell_n)^{r-1})$$

is strictly Lorentzian. f is called the volume polynomial.

In particular, for nef divisors $\ell_1, \dots, \ell_{r-1}$, we have $\deg(\ell_1 \cdots \ell_{r-1}) \geq 0$.

Theorem 5.13. [HX24, Theorem 2.5] *Let f be a degree d Lorentzian polynomial with n variables. Then for any $x \in \mathbb{R}_{\geq 0}^n$ and for any $\alpha, \beta, \gamma \in \mathbb{N}^n$ satisfying $\alpha = \beta + \gamma$, $|\alpha| \leq d$, we have*

$$f(x) \partial^\alpha f(x) \leq c_{\alpha, \beta, \gamma, d} \partial^\beta f(x) \partial^\gamma f(x)$$

for a positive constant $c_{\alpha, \beta, \gamma, d}$ determined by α, β, γ , and d .

(The constant is described in the paper.) Differentiating the volume polynomial with respect to x_i is like intersecting with ℓ_i , by the chain rule. In particular, using the fact that nef divisors are the limit of ample divisors, we have the following.

Corollary 5.14. *Let M be a matroid of rank r , and let m, n, k be nonnegative integers such that $r - 1 \geq k = m + n$. For any combinatorially nef divisors $\ell, \ell_1, \dots, \ell_k$, there exists a constant $c_{m, n, r} > 0$ that only depends on m, n, r , such that*

$$\deg(\ell^{r-1}) \deg(\ell^{r-1-k} \ell_1 \cdots \ell_k) \leq c_{m, n, r} \deg(\ell^{r-1-m} \ell_1 \cdots \ell_m) \deg(\ell^{r-1-n} \ell_{m+1} \cdots \ell_k).$$

For a smooth projective variety X , after replacing the notion ‘combinatorially nef’ with ‘nef’, the inequality in Corollary 5.14 is called the reverse Khovanskii-Teissier (rKT) inequality for a better constant $c_{m, n, r} = \binom{m+n}{m}$.

Proof of Theorem 5.11. We prove the equivalences by a series of implications.

(4) $\Rightarrow$ (1). Pick an ample A . Since D is inside the interior of $\mathcal{F}_M$, $D - \epsilon A$ is fake effective for ϵ small enough.

(1) $\Rightarrow$ (4). Write $D = A + Z$ where A is ample and Z is fake effective. For a divisor B and a small enough ϵ , $D - \epsilon B = (A - \epsilon B) + Z$ is (ample) + (fake effective), which is fake effective.

(1) $\Rightarrow$ (2). Since $D = A + Z$, we have $kD - Z = (k-1)D + A$. Because D is nef and A is ample, this sum is ample.

(2) $\Rightarrow$ (1). $D - \frac{1}{k}Z$ is ample immediately implies D can be written as (ample) + (fake effective).

(3) $\Rightarrow$ (4). Suppose D satisfies (3), and let A be an ample divisor.

The inequality in Corollary 5.14 tells us that for any nef divisors $\ell_2, \dots, \ell_{r-2}$:

$$\deg(D^{r-1}) \deg(A \ell_2 \cdots \ell_{r-2}) \leq c \deg(D^{r-2} A) \deg(D \ell_2 \cdots \ell_{r-2}).$$

After scaling A (for example let $\deg(D^{r-2} A) = 1$), the inequality means $\deg(D \ell_2 \cdots \ell_{r-2}) \geq C_2 \deg(A \ell_2 \cdots \ell_{r-2})$ is true for all nef $\ell_2, \dots, \ell_{r-2}$. Therefore, $D - \epsilon A$ is inside the fake effective cone for small enough ϵ , and D satisfies (4).

(4) $\Rightarrow$ (3). Suppose D is nef and in the interior of the fake effective cone. Then, after scaling there exists an ample divisor A such that $(D - rA)$ is fake effective and $(D + A)$ is ample, where r is the rank of the matroid. By definition $\deg((D - rA)(D + A)^{r-2}) \geq 0$. Therefore,

$$\begin{aligned} 0 &\leq \deg((D - rA)(D + A)^{r-2}) = \deg((D - rA)(D^{r-2} + (r-2)D^{r-3}A + \cdots + A^{r-2})) \\ &= \deg(D^{r-1} - 2D^{r-2}A - \cdots - rA^{r-1}) \end{aligned}$$

That means $\deg(D^{r-1}) > 0$, and D satisfies (3).

□

Remark 5.15. Similarly, for a smooth projective variety X we can define its fake effective cone

$$\mathcal{F} = \{ D \in A^1(X)_{\mathbb{R}} : \deg(D \cdot \ell_1 \cdots \ell_{r-2}) \geq 0 \text{ for all nef } \ell_1, \dots, \ell_{r-2} \}.$$

Every $\mathbb{Q}$ -effective divisor is inside $\mathcal{F}$, but $\mathcal{F}$ can be strictly larger than the actual effective cone. For example, let X be the blow-up of $\mathbb{P}^3$ at four general points, and consider the divisor $D = H - E_1 - E_2 - E_3 - E_4$, where H is the pullback of the hyperplane class and the E_i are the exceptional divisors. We claim that D is in $\mathcal{F}$ but is not $\mathbb{Q}$ -effective.

If mD is effective for a positive integer m , we have a homogeneous degree m polynomial $f(x_1, x_2, x_3, x_4)$ having multiplicity at least m at each of the four points, and we may assume that they are the coordinate vertices. For f to have multiplicity m at $[1 : 0 : 0 : 0]$, $f(1, x_2, x_3, x_4)$ must consist only of terms of degree $\geq m$, and hence x_1 does not appear in f at all. Applying this to the other three vertices forces x_2, x_3 , and x_4 to be absent, which is impossible. Therefore, D is not $\mathbb{Q}$ -effective.

However, $\deg(D \cdot \ell_1 \cdot \ell_2) \geq 0$ for any two nef divisors ℓ_1, ℓ_2 . Write $\ell_k = a_k H - \sum_{i=1}^4 b_{k,i} E_i$. Because ℓ_k is nef, its intersection with a general line in X and with a line in $E_i \cong \mathbb{P}^2$ must be nonnegative, which implies $a_k \geq 0$ and $b_{k,i} \geq 0$. Furthermore, ℓ_k must intersect the strict transform of the line through any two exceptional points nonnegatively, yielding $a_k \geq b_{k,i} + b_{k,j}$ for all $i \neq j$.

The intersection product evaluates to $\deg(D \cdot \ell_1 \cdot \ell_2) = a_1 a_2 - \sum_{i=1}^4 b_{1,i} b_{2,i}$. Assuming that $b_{1,1} \geq b_{1,2} \geq b_{1,3} \geq b_{1,4} \geq 0$, we can bound the sum:

$$\sum_{i=1}^4 b_{1,i} b_{2,i} \leq b_{1,1}(b_{2,1} + b_{2,2}) + b_{1,3}(b_{2,3} + b_{2,4}) \leq (b_{1,1} + b_{1,3})a_2 \leq a_1 a_2.$$

Thus, the intersection is nonnegative, and D is fake effective.

For a smooth projective variety X , the equivalence of (1), (2), (3), and (4) in Goal 5.8 still holds for the fake effective cone via the same proof. Therefore, the interior of the fake effective cone and the interior of the effective cone coincide when restricting to the nef case.

Hence, the “correct” definition of the effective divisors for matroids should strictly contain the cone generated by the x_F ’s (cf. Example 5.9), and be contained inside the fake effective cone.

5.3. Matroid Kawamata–Viehweg vanishing conjecture. Let X be a smooth projective variety of dimension d over $\mathbb{C}$, and let $\mathcal{L}$ be a line bundle on X . If $\mathcal{L}$ is big and nef, the Kawamata–Viehweg vanishing theorem tells us $H^i(X, \mathcal{L}^{-1}) = 0$ for $i < d$. In particular,

$$(-1)^d \chi(X, \mathcal{L}^{-1}) \geq 0.$$

This motivates the following conjecture.

Conjecture 5.16 (Matroid Kawamata–Viehweg Vanishing). *Let M be a matroid of rank r , and let ℓ be a big and nef divisor. If $[\mathcal{L}] \in K(M)$ is the line bundle with $c_1([\mathcal{L}]) = \ell$, then*

$$(-1)^{r-1} \chi(-[\mathcal{L}]) \geq 0.$$

By Proposition 5.5, the conjecture automatically holds when M is realizable over $\mathbb{C}$. Furthermore, it is shown in [Xie10] that the Kawamata–Viehweg vanishing theorem holds for rational surfaces of characteristic p . Hence, Conjecture 5.16 is true when $r = 3$ and the matroid is realizable. This gives us more evidence that the conjecture could be true for general matroids.

In [EL23], the authors proved the following result (they proved something stronger).

Theorem 5.17. [EL23, Theorem 1.5] *Let M be a matroid and $[\mathcal{L}] = \sum_F c_F [\mathcal{L}_F] \in K(M)$ be a line bundle where the c_F are nonnegative integers (cf. Section 2.4). Let d be the numerical dimension of $c_1([\mathcal{L}])$ (i.e., the largest integer t such that $c_1([\mathcal{L}])^t \neq 0$ in the Chow ring). Then*

$$(-1)^d \chi(-[\mathcal{L}]) \geq 0.$$

One might expect Theorem 5.17 to be true for all line bundles $[\mathcal{L}]$ when $c_1([\mathcal{L}])$ is nef. However, the inequality can fail for general classes even when $c_1([\mathcal{L}])$ satisfies (P2). We record a family of counterexamples.

Example 5.18. Let $M = U_{3,2k}$ and write $I = \{1, \dots, k\}$, $J = \{k+1, \dots, 2k\}$. Consider

$$\ell = k\alpha - \sum_{i \in I, j \in J} x_{ij}.$$

One checks that $\ell^2 = 0$, so the numerical dimension of ℓ is 1. Letting $[\mathcal{L}]$ be the line bundle with $c_1([\mathcal{L}]) = \ell$, one obtains

$$\chi(-[\mathcal{L}]) = \deg((1 - \ell) \text{Todd}_M) = \frac{k^2 - 3k + 2}{2}.$$

Hence for $k > 3$ we have $(-1)^1 \chi(-[\mathcal{L}]) = -\frac{k^2 - 3k + 2}{2} < 0$, so the expected sign property fails for this divisor even though ℓ satisfies (P2).

Fix a matroid M of rank r . If ℓ is a nef divisor, then $\alpha + \ell$ is big and nef. Let $[\mathcal{L}]$ be the line bundle with $c_1([\mathcal{L}]) = \ell$. Conjecture 5.16 predicts that

$$(-1)^{r-1} \chi(-[\mathcal{L}] - [\mathcal{L}_E]) \geq 0.$$

Recalling (1), we may compute

$$\chi(-[\mathcal{L}] - [\mathcal{L}_E]) = \deg(\zeta_M(-[\mathcal{L}])\zeta_M(-[\mathcal{L}_E])(1 + \alpha + \alpha^2 + \dots)) = \deg(\zeta_M(-[\mathcal{L}])),$$

as $\zeta_M(-[\mathcal{L}_E]) = 1 - \alpha$.

This motivates the following statement.

Conjecture 5.19 (weaker version of Matroid Kawamata–Viehweg Vanishing). *Let M be a rank r matroid, let ℓ be a nef divisor, and let $[\mathcal{L}]$ be the line bundle with $c_1([\mathcal{L}]) = \ell$. Then*

$$(-1)^{r-1} \deg(\zeta_M(-[\mathcal{L}])) \geq 0.$$

Definition 5.20. Fix a matroid M of rank r . For $0 \leq i \leq r-1$, we define the sets

- (1) $N_i \subset A^i(M)$ to be the set of elements $x \in A^i(M)$ that can be expressed as a product of i nef divisors.
- (2) $P_i \subset A^i(M)$ to be the set of elements $x \in A^i(M)$ that can be expressed as a nonnegative linear combination of elements in N_i .

We then define a set of divisors $\mathcal{P} \subset A^1(M)$ to consist of those divisors ℓ such that $(-1)^i$ times the degree- i part of $\zeta_M(-[\mathcal{L}])$ lies in P_i for all i , where $[\mathcal{L}]$ is the line bundle with $c_1([\mathcal{L}]) = \ell$.

Theorem 5.21. *The set $\mathcal{P}$ satisfies the following properties.*

- (1) $\mathcal{P}$ is contained in the set of nef divisors.
- (2) $\mathcal{P}$ is closed under addition.
- (3) Conjecture 5.19 holds for divisors in $\mathcal{P}$.

Proof. (1) Suppose that $\ell \in \mathcal{P}$ and $[\mathcal{L}]$ is the corresponding line bundle. The degree-1 part of $\zeta_M([\mathcal{L}_F])$ is α_F , and every divisor is a $\mathbb{Z}$ -linear combination of α_F 's. Therefore, the degree-1 part of $\zeta_M([\mathcal{L}])$ is precisely $\ell = c_1([\mathcal{L}])$. Thus, ℓ is a nonnegative linear combination of nef divisors, which is nef.

- (2) If $\ell_1, \ell_2 \in \mathcal{P}$ with $[\mathcal{L}_1], [\mathcal{L}_2]$ being the corresponding line bundles, then $\zeta_M(-[\mathcal{L}_1] - [\mathcal{L}_2]) = \zeta_M(-[\mathcal{L}_1]) \cdot \zeta_M(-[\mathcal{L}_2])$, and $\ell_1 + \ell_2 \in \mathcal{P}$.
- (3) If $\ell \in \mathcal{P}$ with $[\mathcal{L}]$ being the corresponding line bundle, then $(-1)^{r-1} \deg(\zeta_M(-[\mathcal{L}]))$ is a nonnegative linear combination of products of nef divisors, and therefore nonnegative. $\square$

It is natural to ask the following question.

Question 5.22. *When does a nef divisor belong to $\mathcal{P}$?*

In fact, the question is resolved for rank 3 matroids. Let M be a rank 3 matroid, and ℓ be a nef divisor with the corresponding line bundle $[\mathcal{L}]$. Since the degree-1 part of $\zeta_M([\mathcal{L}])$ is ℓ , we only need to check the degree-2 part, and $\ell \in \mathcal{P}$ if and only if Conjecture 5.19 holds for ℓ .

Theorem 5.23. *Let M be a rank 3 matroid. Then $\mathcal{P}$ is exactly the set of nef divisors. Equivalently, Conjecture 5.19 holds for rank 3 matroids.*

The rest of the section is devoted to proving Theorem 5.23.

Proposition 5.24. *Let M be a rank 3 matroid, and let $\ell \in A^1(M)$ be a divisor with $[\mathcal{L}]$ the corresponding line bundle. Then*

$$\zeta_M([\mathcal{L}]) = 1 + \ell + \frac{\ell(\ell + \alpha - S_{1,M})}{2}.$$

(cf. Definition 3.2)

Proof. The formula is verified directly for $[\mathcal{L}_F]$ and $-[\mathcal{L}_F]$. Since it equals $e^{(1+\alpha-S_{1,M})\ell}$ and is multiplicative, it extends to all integral linear combinations of $\pm\alpha_F$, which generate the group of divisors. $\square$

Thus, it remains to show that $\deg(\ell(\ell - \alpha + S_{1,M})) \geq 0$ for a nef divisor ℓ .

Lemma 5.25. *Let M be a matroid of rank r , and suppose $\ell_1, \ell_2, \dots, \ell_k$ are combinatorially nef divisors with $k \leq r-1$. If $\deg(\ell_1^{r-1}) > 0$, then $\ell_1\ell_2\cdots\ell_k = 0$ implies $\ell_2\cdots\ell_k = 0$.*

Proof. In Corollary 5.14, if $k = r-1$, we have

$$\deg(\ell_1^{r-1}) \deg(\ell_1\cdots\ell_{r-1}) \leq c_{m,n,r} \deg(\ell_1^{r-1-m}\ell_1\cdots\ell_m) \deg(\ell_1^{r-1-n}\ell_{m+1}\cdots\ell_{r-1}).$$

Suppose ℓ is a big and nef divisor, and $\deg(\ell_1^{r-1-m}\ell_1\cdots\ell_m) = 0$. This would force $\deg(\ell_1\cdots\ell_{r-1}) = 0$ for all nef divisors $\ell_{m+1}\cdots\ell_{r-1}$. Since the ample cone is nonempty and open, every divisor can be written as the difference of two ample divisors, and we conclude that $\ell_1\cdots\ell_m = 0$. $\square$

Lemma 5.26. *Let M be a rank 3 matroid, and let $\ell \in A^1(M)$ be a nonzero nef divisor. Then there exist a positive integer a , a nonnegative integer n , positive integers $b_1, \dots, b_n$, and rank 2 flats $F_1, \dots, F_n$ such that*

$$\ell = a\alpha - \sum_{i=1}^n b_i x_{F_i}.$$

Proof. The divisor α is big and nef. By Lemma 5.25, we have $\deg(\alpha \cdot \ell) \in \mathbb{Z}_{>0}$. For two rank 1 flats F_i, F_j , the difference $x_{F_i} - x_{F_j}$ can be expressed as a linear combination of x_F where F ranges over rank 2 flats. Hence, $\ell - \deg(\alpha \cdot \ell)\alpha$ is a linear combination of x_F with F of rank 2. Setting $a = \deg(\alpha \cdot \ell)$, we obtain

$$\ell = a\alpha - \sum_{i=1}^n b_i x_{F_i}.$$

Moreover, $\deg(\ell \cdot x_{F_i}) = b_i \geq 0$, which completes the proof. $\square$

Proof of Theorem 5.23. Let $\ell = a\alpha - \sum_{i=1}^n b_i x_{F_i}$ be a nef divisor. We want to show

$$\deg(\ell(\ell - \alpha + S_{1,M})) \geq 0.$$

We compute

$$\deg(\ell^2) = a^2 - \sum_{i=1}^n b_i^2, \quad \deg(\ell \cdot \alpha) = a, \quad \deg(\ell \cdot S_{1,M}) = \sum_{i=1}^n b_i.$$

Therefore,

$$\deg(\ell(\ell - \alpha + S_{1,M})) = a(a-1) - \sum_{i=1}^n b_i(b_i-1).$$

Since ℓ is nef, we have $\deg(\ell^2) = a^2 - \sum_{i=1}^n b_i^2 \geq 0$. This inequality implies

$$a(a-1) - \sum_{i=1}^n b_i(b_i-1) \geq 0.$$

Indeed, by adjoining additional 1's to the sequence (b_i) (which do not affect $b_i(b_i-1)$), we may assume $a^2 = \sum_{i=1}^n b_i^2$. In this case, $a \leq \sum_{i=1}^n b_i$, which yields the desired inequality. $\square$

In summary, we have verified Conjecture 5.19 for rank 3 matroids. For the stronger Conjecture 5.16, however, the inequality becomes

$$(a-1)(a-2) - \sum_{i=1}^n b_i(b_i-1) \geq 0,$$

which need not hold without incorporating the combinatorial structure of the matroid. To the author's knowledge, this conjecture remains open even in rank 3.

Remark 5.27. The same proof shows that Lemma 5.25 generalizes to smooth projective varieties. This result also yields interesting consequences in the matroid setting. Let M be a matroid of rank r , and let $\ell_1, \dots, \ell_k$ be nef divisors with $k < r$. Because α and β are big and nef, if $\ell_1 \cdots \ell_k \neq 0$, then $\deg(\alpha^{r-i-k} \beta^{i-1} \ell_1 \cdots \ell_k) > 0$ for any $1 \leq i \leq r-k$. Consequently, if we express the non-zero product $\ell_1 \cdots \ell_k$ as a nonnegative linear combination of $x_{F_1} \cdots x_{F_k}$, there must exist a flag in $\mathcal{H}_{\{i, \dots, i+k-1\}}$ (cf. Definition 3.11) with a strictly positive coefficient. This naturally raises the question: what types of flags of flats are guaranteed to appear in products of nef divisors?

6. PROPERTIES OF THE β CLASSES

While the α_F classes corresponding to flats are fundamental to the intersection theory of matroids, they represent only one part of the geometric picture. In this section, we introduce the β_S classes—associated with arbitrary non-empty subsets S of the ground set E —and establish their core properties as natural geometric and combinatorial analogues to the α_F classes.

We structure our investigation into three main areas:

- **Exceptional isomorphisms:** We analyze the behavior of β_S under the exceptional isomorphism ζ_M .
- **Numerical dimensions:** We compute the numerical dimensions of the β_S classes, establishing that intersection products of β classes are strictly positive if and only if their corresponding subsets satisfy the dragon-Hall-Rado condition.
- **Euler characteristics and positivity:** We investigate the Euler characteristics of β classes and prove specific cases of Conjecture 6.10.

6.1. Exceptional isomorphism for β classes. The classes $[\mathcal{L}_F] \in K(M)$ correspond to divisors α_F , and the exceptional isomorphism ζ_M sends $[\mathcal{L}_F]$ to $1 + \alpha_F + \alpha_F^2 + \cdots$. We first establish the analogous statement for the β_S classes.

Definition 6.1. Let M be a matroid on the ground set E . For a nonempty subset $S \subseteq E$, let $[\mathcal{K}_S] \in K(M)$ denote the line bundle with $c_1([\mathcal{K}_S]) = \beta_S$.

Recall that for a matroid M on a ground set E and an element $i \in E$, there is a deletion map (Lemma 2.9) $\theta_i : A^*(M \setminus i) \rightarrow A^*(M)$ given by the projection map between the toric varieties (see [BHMPW22, Proposition 3.1]). Therefore, there is also a deletion map $\tilde{\theta}_i : K(M \setminus i) \rightarrow K(M)$

between the K -rings, and the map commutes with the Chern class map and θ_i (see [Ful98, Theorem 3.2(d)]):

$$\begin{array}{ccc} K(M \setminus i) & \xrightarrow{\tilde{\theta}_i} & K(M) \\ c_t \downarrow & & \downarrow c_t \\ A^*(M \setminus i) & \xrightarrow{\theta_i} & A^*(M) \end{array}$$

Lemma 6.2. *Let M be a matroid on the ground set E , and let $i \in E$. The deletion map θ_i commutes with the ζ map. More precisely, we have:*

$$\zeta_M \circ \tilde{\theta}_i = \theta_i \circ \zeta_{M \setminus i}.$$

Proof. Because the line bundle classes $[\mathcal{L}_F]$ generate the K -ring (cf. Section 2.4), it suffices to check that $\zeta_M \circ \tilde{\theta}_i([\mathcal{L}_F]) = \theta_i \circ \zeta_{M \setminus i}([\mathcal{L}_F])$ for any flat F in $M \setminus i$.

For a flat F in $M \setminus i$, $\theta_i(\alpha_F)$ is exactly $\alpha_{\text{cl}(F)}$ in M . Since $\tilde{\theta}_i$ commutes with the Chern class map, we conclude that $\tilde{\theta}_i([\mathcal{L}_F]) = [\mathcal{L}_{\text{cl}(F)}]$. Therefore, the lemma follows from the fact that $\zeta_{M \setminus i}([\mathcal{L}_F]) = 1 + \alpha_F + \alpha_F^2 + \dots$. $\square$

In [BEST23, Theorem 10.11], it is shown that ζ_M sends $[\mathcal{K}_E]$ to $1 + \beta_E$. We show the analogue for β_S .

Lemma 6.3. *Let M be a matroid on the ground set E and let $S \subseteq E$ be a nonempty subset. Let $S^c = E \setminus S = \{s_1, \dots, s_k\}$. Then:*

$$\beta_S = \theta_{S^c}(\beta_{M \setminus S^c}) := \theta_{s_1} \circ \dots \circ \theta_{s_k}(\beta_{M \setminus S^c}).$$

Note that β_S lives in $A^(M)$, while $\beta_{M \setminus S^c}$ is the β class of the matroid $M \setminus S^c$.*

Proof. We pick $j \in E \setminus S^c$ and suppose that $\beta_{M \setminus S^c}$ is the sum of flats in $M \setminus S^c$ that do not contain j . Through the sequence of θ maps, every flat that does not contain j and is not contained in S^c will appear, which exactly matches the definition of β_S . $\square$

Theorem 6.4. *For nonempty subsets $S \subseteq E$, ζ_M sends $[\mathcal{K}_S]$ to $1 + \beta_S$.*

Proof. Let $[\mathcal{K}_{M \setminus S^c}] \in K(M \setminus S^c)$ denote the line bundle in $M \setminus S^c$ corresponding to its $\beta = \beta_E$ class. Since $\zeta_{M \setminus S^c}([\mathcal{K}_{M \setminus S^c}]) = 1 + \beta_{M \setminus S^c}$, we conclude that $\zeta_M([\mathcal{K}_S]) = 1 + \beta_S$ by Lemma 6.2 and Lemma 6.3. $\square$

Corollary 6.5. *For nonempty subsets $S \subseteq E$, the divisors α_S, β_S lie in $\mathcal{P}$ (cf. Definition 5.20). In particular, Conjecture 5.19 holds for a positive (integral) linear combination of α_S and β_S .*

Proof. We have $\zeta_M(-[\mathcal{L}_S]) = 1 - \alpha_S$, and $\alpha_S \in \mathcal{P}$. Theorem 6.4 shows that $\zeta_M(-[\mathcal{K}_S]) = 1 - \beta_S + \beta_S^2 - \dots$, and $\beta_S \in \mathcal{P}$. $\square$

Corollary 6.6. *Let $S_1, \dots, S_j$ and $T_1, \dots, T_k$ be subsets of E . If $\sum_{i=1}^j \beta_{S_i} = \sum_{i=1}^k \beta_{T_i}$, then*

$$\prod_{i=1}^j (1 + \beta_{S_i}) = \prod_{i=1}^k (1 + \beta_{T_i}).$$

Proof. The equation $\sum_{i=1}^j \beta_{S_i} = \sum_{i=1}^k \beta_{T_i}$ implies $\sum_{i=1}^j [\mathcal{K}_{S_i}] = \sum_{i=1}^k [\mathcal{K}_{T_i}]$. Applying ζ_M on both sides concludes the proof. $\square$

This identity also holds for the α_F 's; however, this is less interesting because $\alpha_F = 0$ if F is a rank-1 flat, and the set $\{\alpha_F : F \text{ is a rank } r > 1 \text{ flat}\}$ forms a basis of $A^1(M)$.

Corollary 6.7. Let $S_1, \dots, S_j$ and $T_1, \dots, T_k$ be subsets of E . Let td_i denote the degree- i component of the Todd class $\text{td}(T_M)$. Then

$$\deg\left(\prod_{i=1}^j \alpha_{S_i} \prod_{i=1}^k \beta_{T_i} \text{td}_{r-j-k-1}\right) = \deg\left((-1)^j \prod_{i=1}^j \log(1 - \alpha_{S_i}) \prod_{i=1}^k \log(1 + \beta_{T_i}) (1 + \alpha + \alpha^2 + \dots)\right).$$

Proof. Compare the coefficient of $s_1 \cdots s_j t_1 \cdots t_k$ in

$$\chi\left(\sum_{i=1}^j s_i [\mathcal{L}_{S_i}] + \sum_{i=1}^k t_i [\mathcal{K}_{T_i}]\right)$$

as given by Proposition 2.6 and Equation (1). $\square$

Corollary 6.8. For a proper flat F of M , let $[\mathcal{X}_F] \in K(M)$ denote the line bundle with $c_1([\mathcal{X}_F]) = x_F$. We have the following.

- (1) If F_1 is a rank 1 flat, then $\zeta_M(-[\mathcal{X}_{F_1}]) = 1 - x_{F_1}$.
- (2) If F_{r-1} is a rank $r-1$ flat, then $\zeta_M([\mathcal{X}_{F_{r-1}}]) = 1 + x_{F_{r-1}}$.

Proof. For (1), one has the identity $-x_{F_1} = -\beta + \beta_{E \setminus F_1}$. Using $\zeta_M([\mathcal{K}_E]) = 1 + \beta$ and $\zeta_M([\mathcal{K}_{E \setminus F_1}]) = 1 + \beta_{E \setminus F_1}$ and the relation $x_{F_1} \beta = 0$, a short algebraic computation yields $\zeta_M(-[\mathcal{X}_{F_1}]) = 1 - x_{F_1}$. The argument for (2) is analogous, where we consider α and $\alpha_{F_{r-1}}$ instead. $\square$

Remark 6.9. We can compute $\zeta_M([\mathcal{X}_F])$ for general flats F using the same method. Although $\zeta_M([\mathcal{X}_{F_1}]) = 1 + x_{F_1} + x_{F_1}^2 + \dots + x_{F_1}^{r-1}$ and $\zeta_M([\mathcal{X}_{F_{r-1}}]) = 1 + x_{F_{r-1}}$, it is *false* that $\zeta_M([\mathcal{X}_{F_{r-2}}]) = 1 + x_{F_{r-2}} + x_{F_{r-2}}^2$.

6.2. Numerical dimension of β classes. It is natural to ask whether Theorem 5.17 extends to positive integral linear combinations of $[\mathcal{L}_S]$ and $[\mathcal{K}_S]$. We formulate the following conjecture:

Conjecture 6.10. Let M be a matroid and let $[\mathcal{L}]$ be a positive integral linear combination of the classes $[\mathcal{L}_S]$ and $[\mathcal{K}_S]$. Let d be the numerical dimension of $c_1([\mathcal{L}])$. Then

$$(-1)^d \chi(-c_1([\mathcal{L}])) \geq 0.$$

The first question is to compute the numerical dimension of β_S .

Definition 6.11. For a matroid M on the ground set E and a sequence $(S_1, \dots, S_m)$ of non-empty subsets of E , we say the sequence satisfies the dragon-Hall-Rado condition (with respect to M) if

$$\text{rk}_M\left(\bigcup_{i \in I} S_i\right) \geq 1 + |I|, \text{ for all } \emptyset \subsetneq I \subseteq [m].$$

For classes α_F , it is shown that

Proposition 6.12. [BES24, Theorem 5.2.4] Let M be a matroid of rank r , and let $F_1, \dots, F_{r-1}$ be flats of M (with repeats allowed). Then

$$\deg(\alpha_{F_1} \cdots \alpha_{F_{r-1}}) = \begin{cases} 1 & \text{if } (F_1, \dots, F_{r-1}) \text{ satisfies the dragon-Hall-Rado condition.} \\ 0 & \text{else.} \end{cases}$$

This proposition remains true even if we drop the requirement that the F_i are flats.

Theorem 6.13. Let M be a matroid of rank r , and let $S_1, \dots, S_{r-1}$ be subsets of E (with repeats allowed). Then:

$$\deg(\beta_{S_1} \cdots \beta_{S_{r-1}}) > 0 \iff \deg(\alpha_{S_1} \cdots \alpha_{S_{r-1}}) > 0.$$

In particular, $\deg(\beta_{S_1} \cdots \beta_{S_{r-1}}) > 0$ if and only if $(S_1, \dots, S_{r-1})$ satisfies the dragon-Hall-Rado condition.

Before proving this theorem, we establish a few preliminary lemmas. For a flag of flats $\mathcal{F} = (F_1 \subsetneq \dots \subsetneq F_k)$, we denote the monomial $x_{F_1} \cdots x_{F_k}$ by $x_{\mathcal{F}}$.

Lemma 6.14. *Let M be a matroid of rank r , and let $\ell_1, \dots, \ell_k$ be nef divisors. The product $\ell_1 \cdots \ell_k$ is a nonnegative linear combination of $x_{\mathcal{F}}$ for flags of flats $\mathcal{F}$.*

Proof. We prove the result by induction on k . The result is trivially true for $k = 1$.

For a nef divisor ℓ and a flag of flats $\mathcal{F} = (F_1 \subsetneq \dots \subsetneq F_k)$, we can write $\ell = \sum c_F x_F$ such that $c_{F_i} = 0$ and $c_F \geq 0$ for any F where $F \cup \mathcal{F}$ forms a valid flag of flats. All other terms will evaluate to zero when multiplied by $x_{\mathcal{F}}$, which completes the induction. $\square$

If we consider ample divisors instead, the product is a strictly positive linear combination of all possible $x_{\mathcal{F}}$.

Corollary 6.15. *Let $D = \sum_F c_F x_F$ with $c_F \geq 0$, and let $\ell_1, \dots, \ell_d$ be nef divisors. Then*

$$D \cdot (\ell_1 \cdots \ell_d) = 0 \iff c_F x_F (\ell_1 \cdots \ell_d) = 0 \text{ for all } F.$$

Proof. $c_F x_F (\ell_1 \cdots \ell_d)$ can be written as a nonnegative linear combination of $x_{\mathcal{F}}$ for flag of flats $\mathcal{F}$.

We may assume $r - d - 2 \geq 0$. Let A be an ample divisor. Then

$$\begin{aligned} \sum c_F x_F \cdot \ell_1 \cdots \ell_d = 0 &\implies \deg(\sum c_F x_F \cdot \ell_1 \cdots \ell_d A^{r-d-2}) = 0 \\ &\implies \deg(c_F x_F \cdot \ell_1 \cdots \ell_d A^{r-d-2}) = 0 \implies c_F x_F \cdot \ell_1 \cdots \ell_d = 0. \end{aligned}$$

$\square$

Proof of Theorem 6.13. We prove the following stronger statement: For a set $S \subset E$ and nef divisors $\ell_1, \dots, \ell_d$, $d < r - 1$, we have

$$\beta_S \cdot \ell_1 \cdots \ell_d \neq 0 \iff \alpha_S \cdot \ell_1 \cdots \ell_d \neq 0.$$

For any $i \in S$, α_S consists of flats containing i but not containing S . Therefore, if we sum over all $i \in S$ and let $n = |S|$, we have

$$\alpha_S = \sum_{j=1}^{n-1} \left(\frac{j}{n} \sum_{|F \cap S|=j} x_F \right).$$

Similarly, for any $i \in S$, β_S consists of flats not containing i but not contained in $E \setminus S$, we have

$$\beta_S = \sum_{j=1}^{n-1} \left(\frac{n-j}{n} \sum_{|F \cap S|=j} x_F \right).$$

Therefore, by Corollary 6.15,

$$\begin{aligned} \beta_S \cdot \ell_1 \cdots \ell_d \neq 0 &\iff x_F \cdot \ell_1 \cdots \ell_d \neq 0 \text{ for a flat } F \text{ such that } 0 < |F \cap S| < n \\ &\iff \alpha_S \cdot \ell_1 \cdots \ell_d \neq 0. \end{aligned}$$

The proof of the original theorem then proceeds by exchanging β_{S_i} and α_{S_i} one by one. $\square$

In particular, the numerical dimension of β_S equals $\text{rk}(S) - 1$.

Question 6.16. *Do we have a simple formula for $\deg(\beta_{S_1} \cdots \beta_{S_{r-1}})$ if it is nonzero?*

Remark 6.17. Let $S = \bigcup_{i=1}^{r-1} S_i$. We will have $\deg(\beta_{S_1} \cdots \beta_{S_{r-1}}) \leq \deg(\beta_S^{r-1})$, but equality may not hold.

The proof also suggests the notion of $\gamma_S = \alpha_S + \beta_S$, which is the sum of x_F where $0 < |F \cap S| < |S|$; this is why we use the notation β_S for subtracting flats contained in $E \setminus S$ instead of in S .

6.3. Euler characteristic for β classes. Finally, in this section, we investigate the Euler characteristics of β classes. For a positive integer n , it is known (see [EL23, Example 5.8]) that

$$(-1)^{r-1}\chi(-n[\mathcal{K}_E]) \geq 0.$$

Corollary 6.18. *Let M be a matroid of rank r on the ground set E , and let $S \subseteq E$ be a nonempty subset. Setting $S^c = E \setminus S$, suppose that $\text{rk}(S) = i$ (so that $i - 1$ is the numerical dimension of β_S). Let $[\mathcal{K}_{M \setminus S^c}] \in K(M \setminus S^c)$ denote the line bundle in $M \setminus S^c$ corresponding to its $\beta = \beta_E$ class. Then for any integer k , we have $\chi(k[\mathcal{K}_S]) = \chi(k[\mathcal{K}_{M \setminus S^c}])$. In particular, for any positive integer n , we have*

$$(-1)^{i-1}\chi(-n[\mathcal{K}_S]) \geq 0.$$

Proof. By Lemma 2.9, the deletion map commutes with the degree map up to multiplication by a power of α , which accounts for any difference in rank between the respective matroids. Furthermore, Lemma 6.3 states that the composition of deletion maps, denoted θ_{S^c} , sends $\beta_{M \setminus S^c} \in A^*(M \setminus S^c)$ to $\beta_S \in A^*(M)$. One can also check that it maps $\alpha_{M \setminus S^c}$ to α_M .

The Euler characteristic can be evaluated by $\chi(k[\mathcal{K}_E]) = \deg((1 + \beta)^k(1 + \alpha + \alpha^2 + \dots))$. Applying the composition θ_{S^c} to this expansion for $\beta_{M \setminus S^c}$ yields $\chi(k[\mathcal{K}_S]) = \chi(k[\mathcal{K}_{M \setminus S^c}])$. $\square$

To conclude the section, we prove a special case of Conjecture 6.10.

Lemma 6.19. *Let $F_1, \dots, F_m$ be rank-1 flats such that each $S_i := E \setminus F_i$ has rank r . Fix positive integers $n, a_1, \dots, a_m$ with $n \geq \sum_{i=1}^m a_i$, and set*

$$B = (n - \sum_{i=1}^m a_i)\beta + \sum_{i=1}^m a_i\beta_{S_i} = n\beta - \sum_{i=1}^m a_i x_{F_i}.$$

Then B has numerical dimension $r - 1$. Let $[\mathcal{B}] \in K(M)$ be the line bundle with $c_1([\mathcal{B}]) = B$. We have

$$(-1)^{r-1}\chi(-[\mathcal{B}]) \geq 0.$$

Sketch of proof. The rank-1 flats x_{F_i} pairwise have vanishing intersection with β (and with each other in the relevant degrees), so mixed products of β and the x_{F_i} vanish. Hence the Chern character of $-\mathcal{B}$ decomposes essentially as a sum of simpler contributions coming from $-n\beta$ and the x_{F_i} ; this leads to the identity

$$\chi(-[\mathcal{B}]) = \chi(-n[\mathcal{K}_E]) - \sum_{i=1}^m \chi(-a_i[\mathcal{K}_E]) + \sum_{i=1}^m \chi(-a_i[\mathcal{K}_{S_i}]).$$

(The displayed equality follows from the vanishing of cross terms together with the relation $\beta_{S_i} = \beta - x_{F_i}$ for rank-1 flats.)

To analyse the first two terms, one uses the deletion-contraction type relation for $\chi(-j[\mathcal{K}_E])$: for any non-loop, non-coloop element $i \in E$ (and assuming the contraction M_i is loopless) one has

$$\chi(M, -j[\mathcal{K}_E]) = \chi(M \setminus i, -j[\mathcal{K}_E]) - \sum_{k=1}^j \chi(M_i, -k[\mathcal{K}_E]).$$

By induction on the size of the ground set (applying this relation and using the sign properties for lower-rank contractions) one shows that, for fixed matroid M of rank $r > 1$, the sequence $\frac{(-1)^{r-1}\chi(M, -n[\mathcal{K}_E])}{n}$ is nondecreasing in n . Consequently

$$(-1)^{r-1}\chi(-n[\mathcal{K}_E]) - \sum_{i=1}^m (-1)^{r-1}\chi(-a_i[\mathcal{K}_E]) \geq 0.$$

Finally, each term $\chi(-a_i[\mathcal{K}_{S_i}])$ satisfies the expected sign condition: $(-1)^{r-1}\chi(-a_i[\mathcal{K}_{S_i}]) \geq 0$. Combining the three displayed inequalities yields $(-1)^{r-1}\chi(-[\mathcal{B}]) \geq 0$, as required. (When $r = 1$ the statement is trivial because the geometry is a point.) $\square$

The more general statement for Theorem 5.17 has recently been proved by Eur, Fink, and Larson for divisors that are pullbacks of nef line bundles from the permutohedral variety X_E ([EFL25, Theorem B]). Consequently, Conjecture 6.10 and several derived statements in Section 6 become automatic. We include a self-contained treatment here, and we also analyze the more general combinatorially nef and big-and-nef cases, which remain open beyond the scope of [EFL25].

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